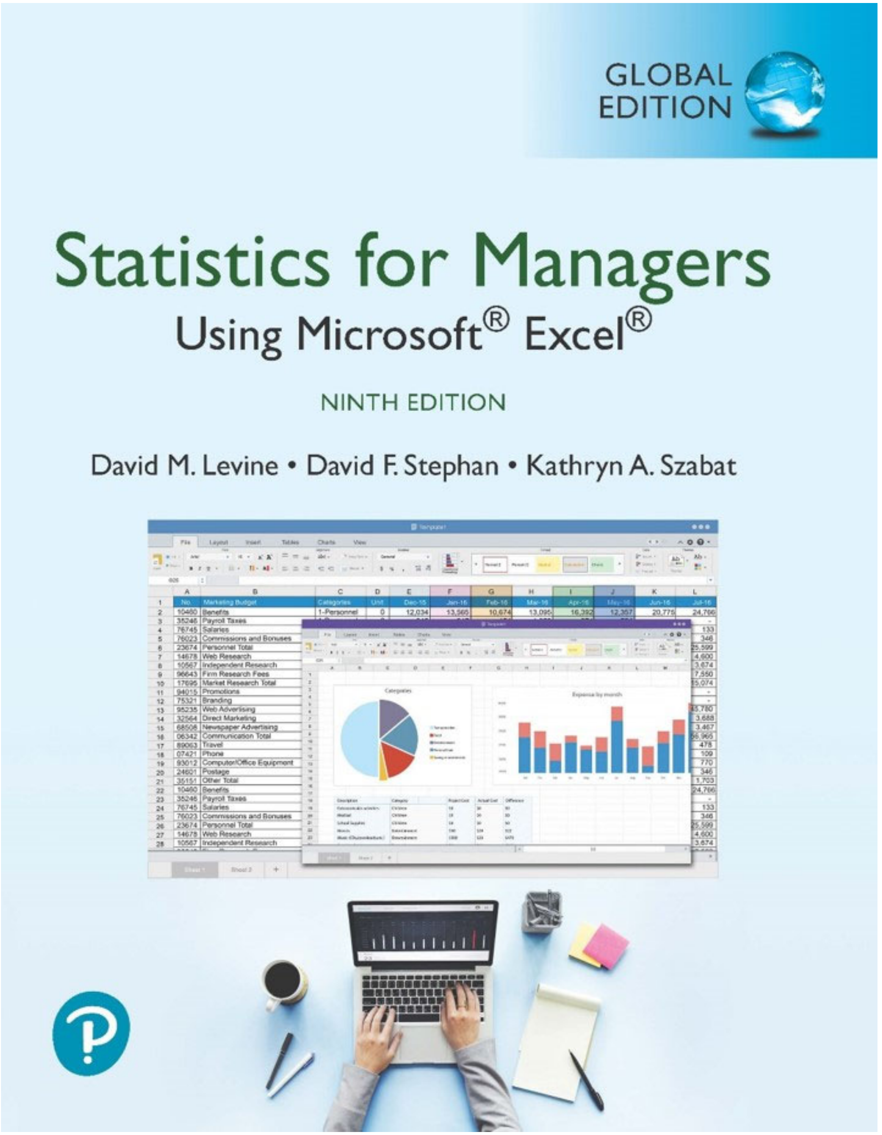


Chapter 6

The normal distribution and other continuous distributions

Objectives

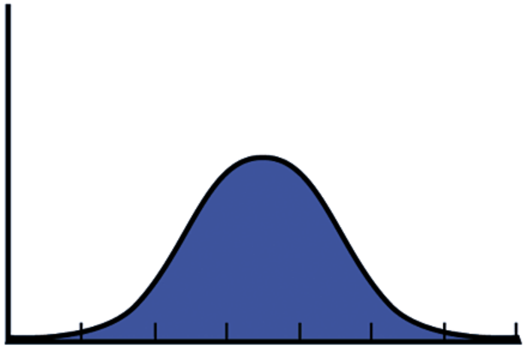
In this chapter, you learn:

- To compute probabilities from the normal distribution.
- How to use the normal distribution to solve business problems.
- To use the normal probability plot to determine whether a data set is approximately normally distributed.
- Compute probabilities from the uniform distribution.
- Compute probabilities from the exponential distribution.

Continuous probability distributions

- A continuous variable is a variable that can assume any value on a continuum:
  - Thickness of an item.
  - Time required to complete a task.
  - Temperature of water.
  - Height of an item.
- These can potentially take on any value depending only on the ability to measure precisely and accurately.

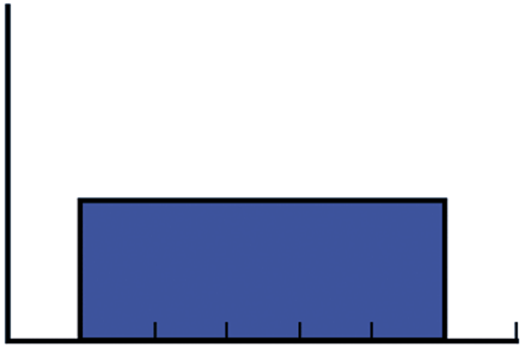
Continuous probability distributions vary by shape



Values of X

Normal Distribution

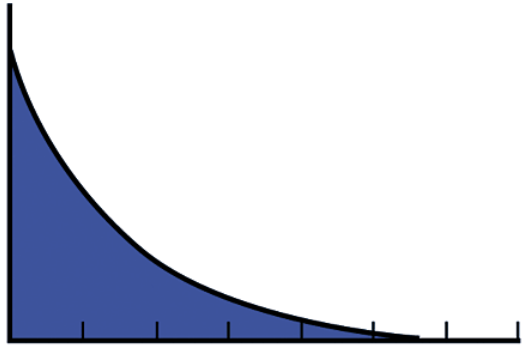
- Symmetrical
- Bell-shaped
- Ranges from negative to positive infinity



Values of X

Uniform Distribution

- Symmetrical
- Also known as Rectangular Distribution
- Every value between the smallest & largest is equally likely



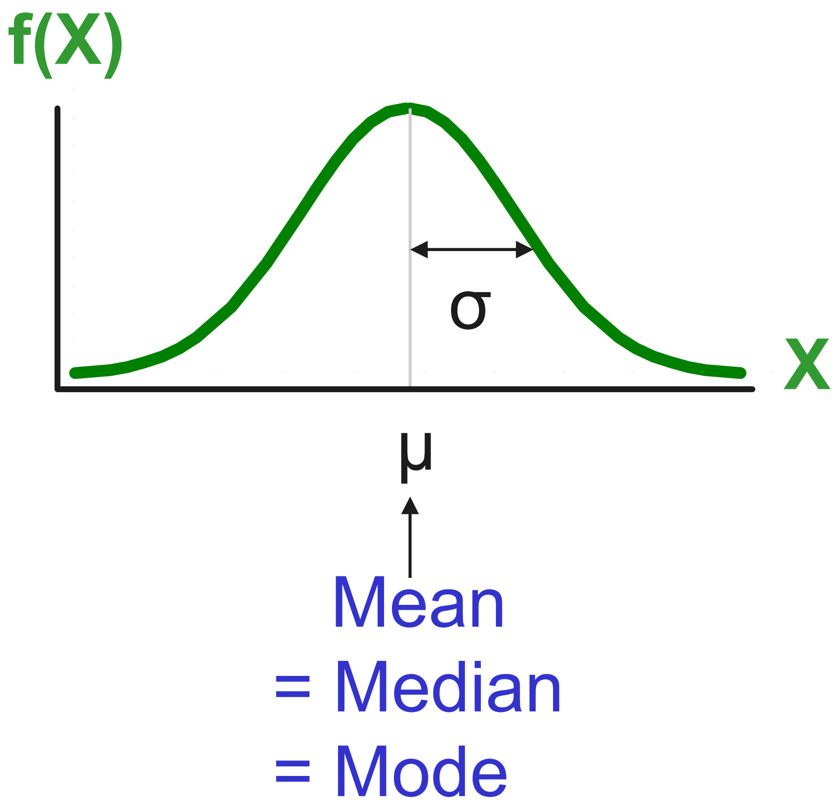
Values of X

Exponential Distribution

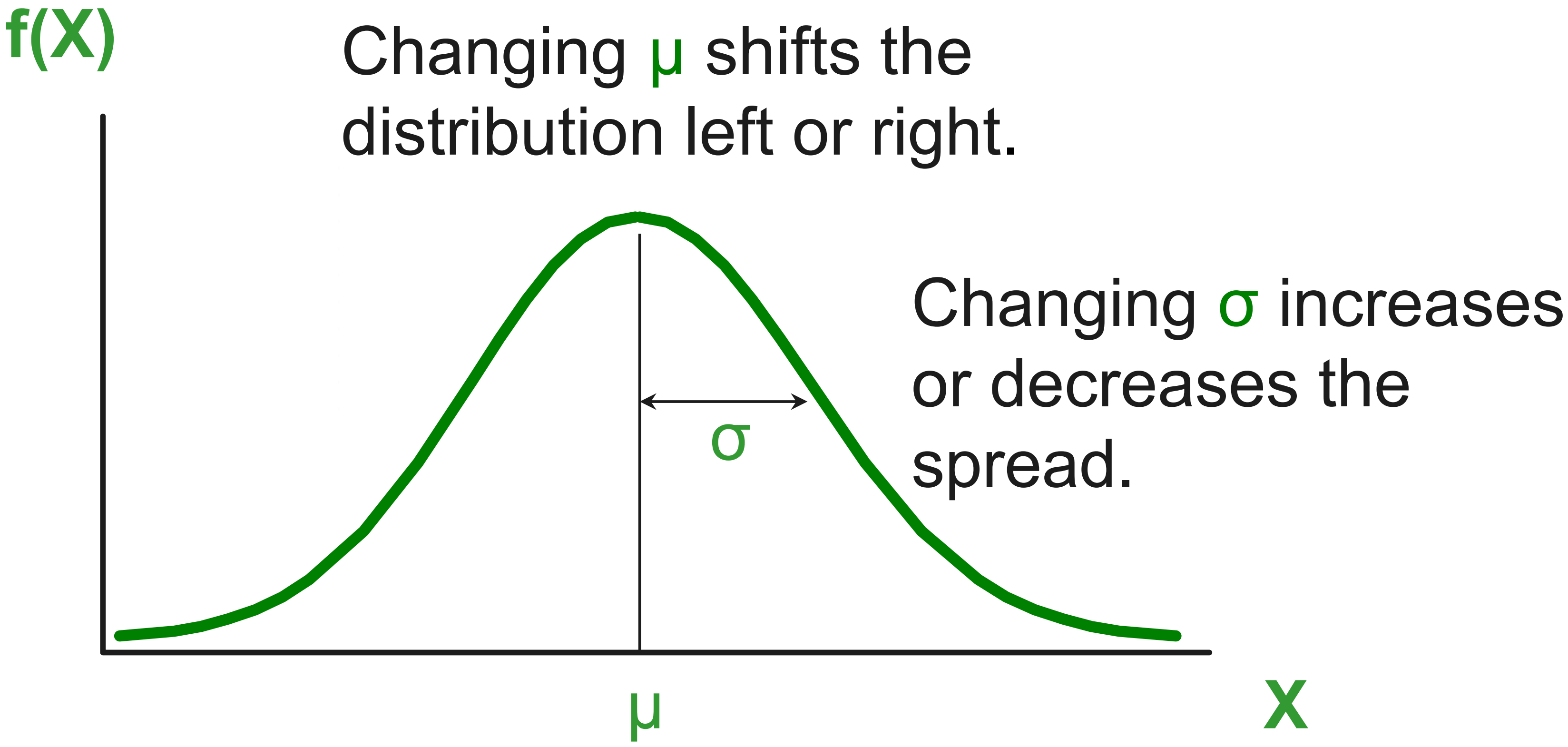
- Right skewed
- Mean > Median
- Ranges from zero to positive infinity

The normal distribution

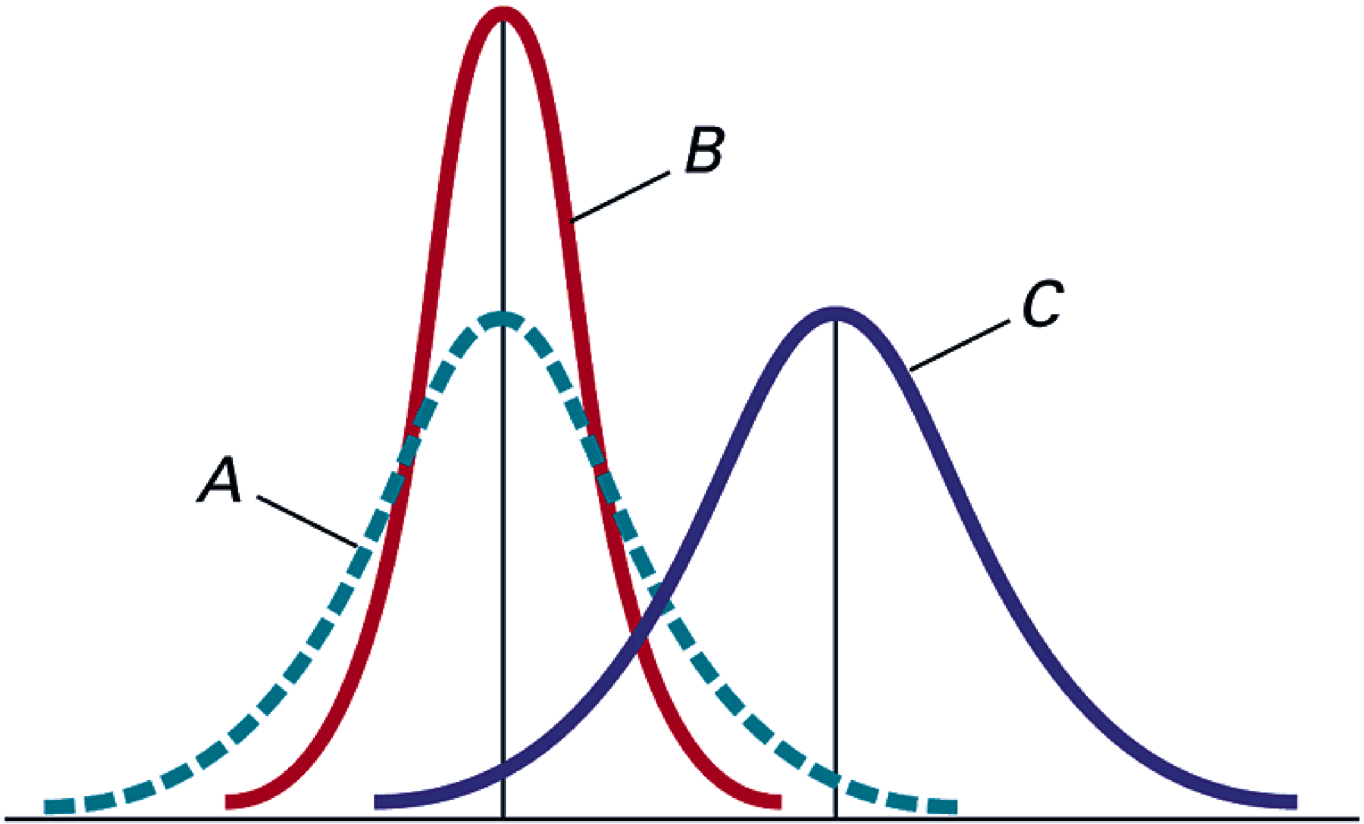
- Bell Shaped.
- Symmetrical.
- Mean = Median = Mode.
- Location is determined by the mean, μ.
- Spread is determined by the standard deviation, σ.
- The random variable has an infinite theoretical range. That is, $-\infty < X < +\infty$.



The normal distribution shape



By varying the parameters μ and σ, we obtain different normal distributions



- A and B have the same mean but different standard deviations.
- A and C have the same standard deviation but different means.
- B and C have different means and different standard deviations.

The normal distribution density function

- The formula for the normal probability density function is:

$$f(X) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{1}{2}\left(\frac{(X-\mu)}{\sigma}\right)^2}$$

where e = the mathematical constant approximated by 2.7183

π = the mathematical constant approximated by 3.1416

μ = the population mean

σ = the population standard deviation

X = any value of the continuous variable

The standardised normal

- Any normal distribution (with any mean and standard deviation combination) can be transformed into the standardised normal distribution (Z).

- To compute normal probabilities, one needs to transform X units into Z units.

- The standardised normal distribution (Z) has a mean of 0 and a standard deviation of 1.

Translation to the standardised normal distribution

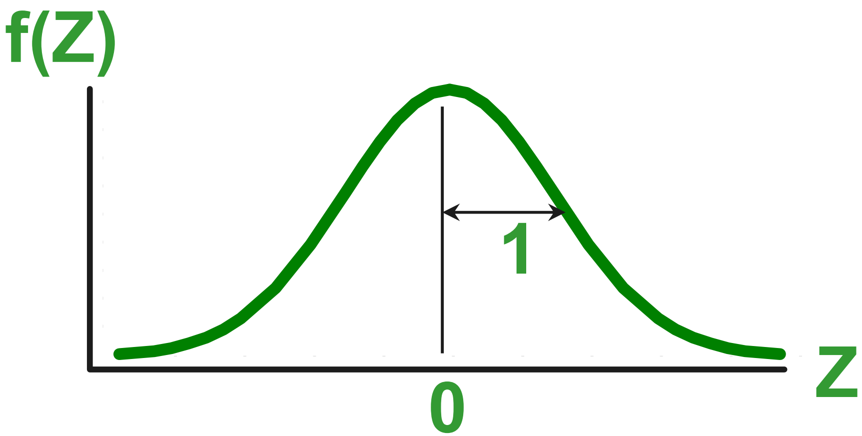
- Translate from X to the standardised normal (the “Z” distribution) by subtracting the mean of X and dividing by its standard deviation:

$$Z = \frac{X - \mu}{\sigma}$$

The Z distribution always has mean = 0 and standard deviation = 1.

The standardised normal distribution

- Also known as the “Z” distribution.
- Mean is 0.
- Standard Deviation is 1.



Values above the mean have positive Z-values.

Values below the mean have negative Z-values.

The standardised normal probability density function

- The formula for the standardised normal probability density function is:

$$f(Z) = \frac{1}{\sqrt{2\pi}} e^{-(1/2)Z^2}$$

where e = the mathematical constant approximated by 2.7183

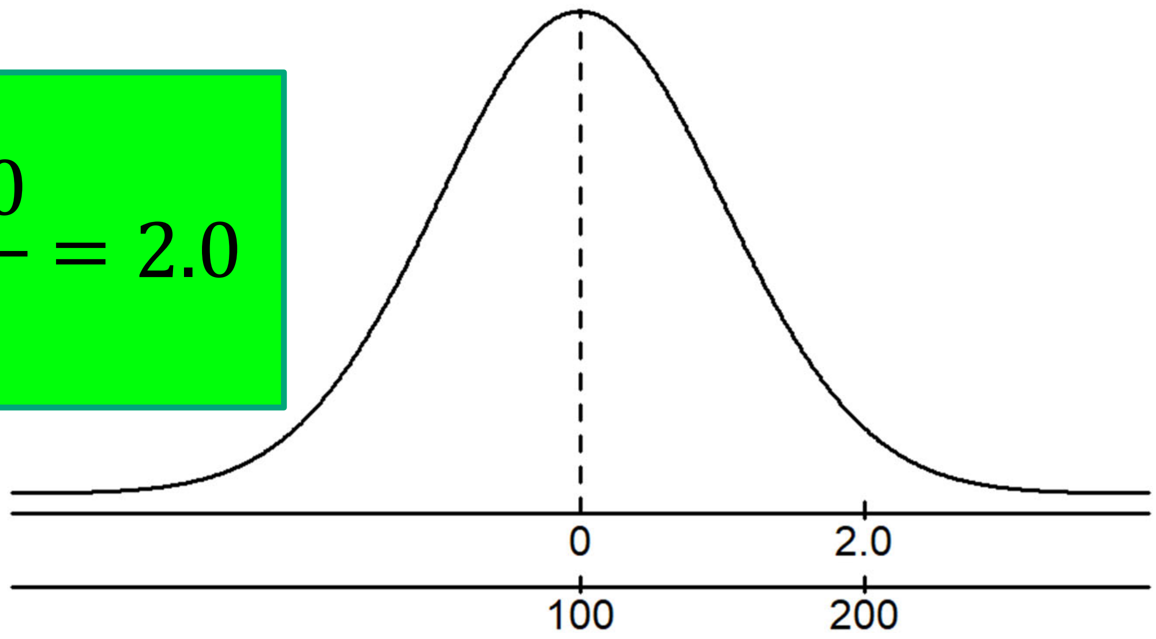
π = the mathematical constant approximated by 3.1416

Z = any value of the standardised normal distribution

Example

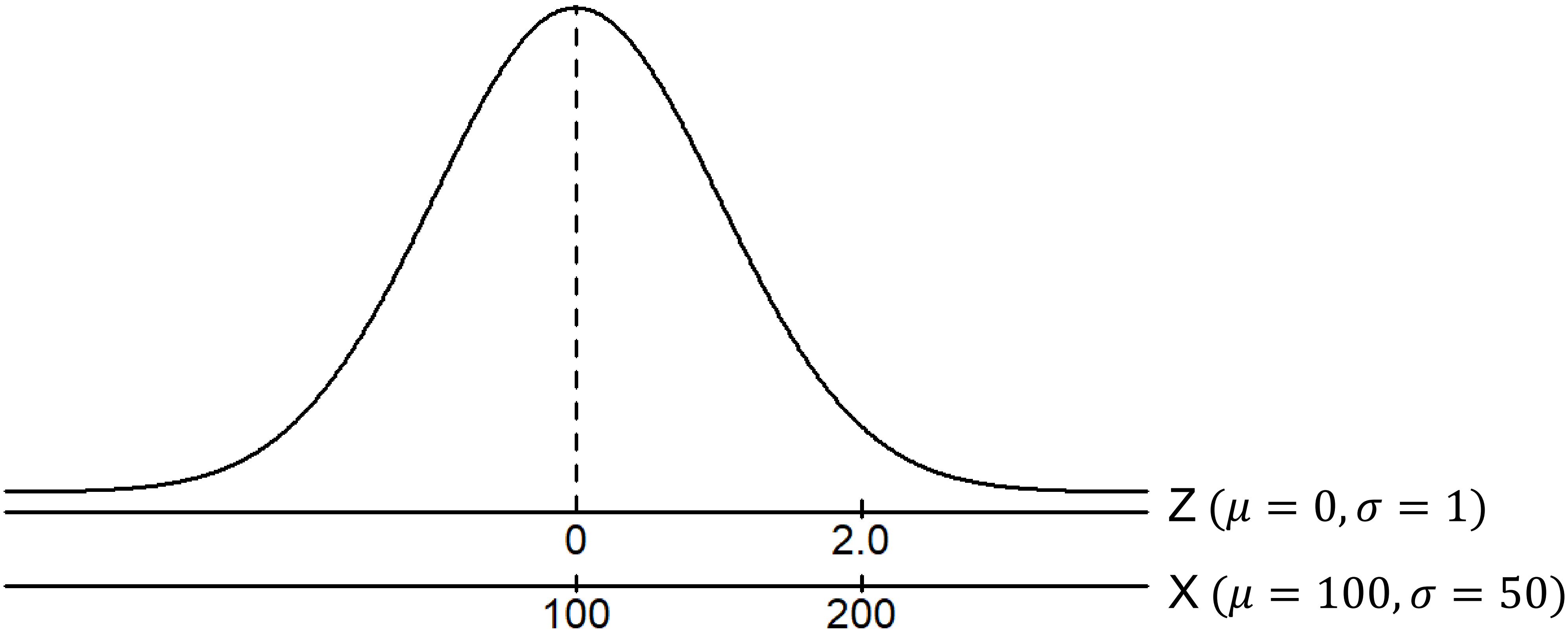
- If X is distributed normally with mean of 100 and standard deviation of 50, the z value for X = 200 is:

$$z = \frac{x - \mu}{\sigma} = \frac{200 - 100}{50} = 2.0$$



- This means that X = 200 is two standard deviations above the mean of 100.

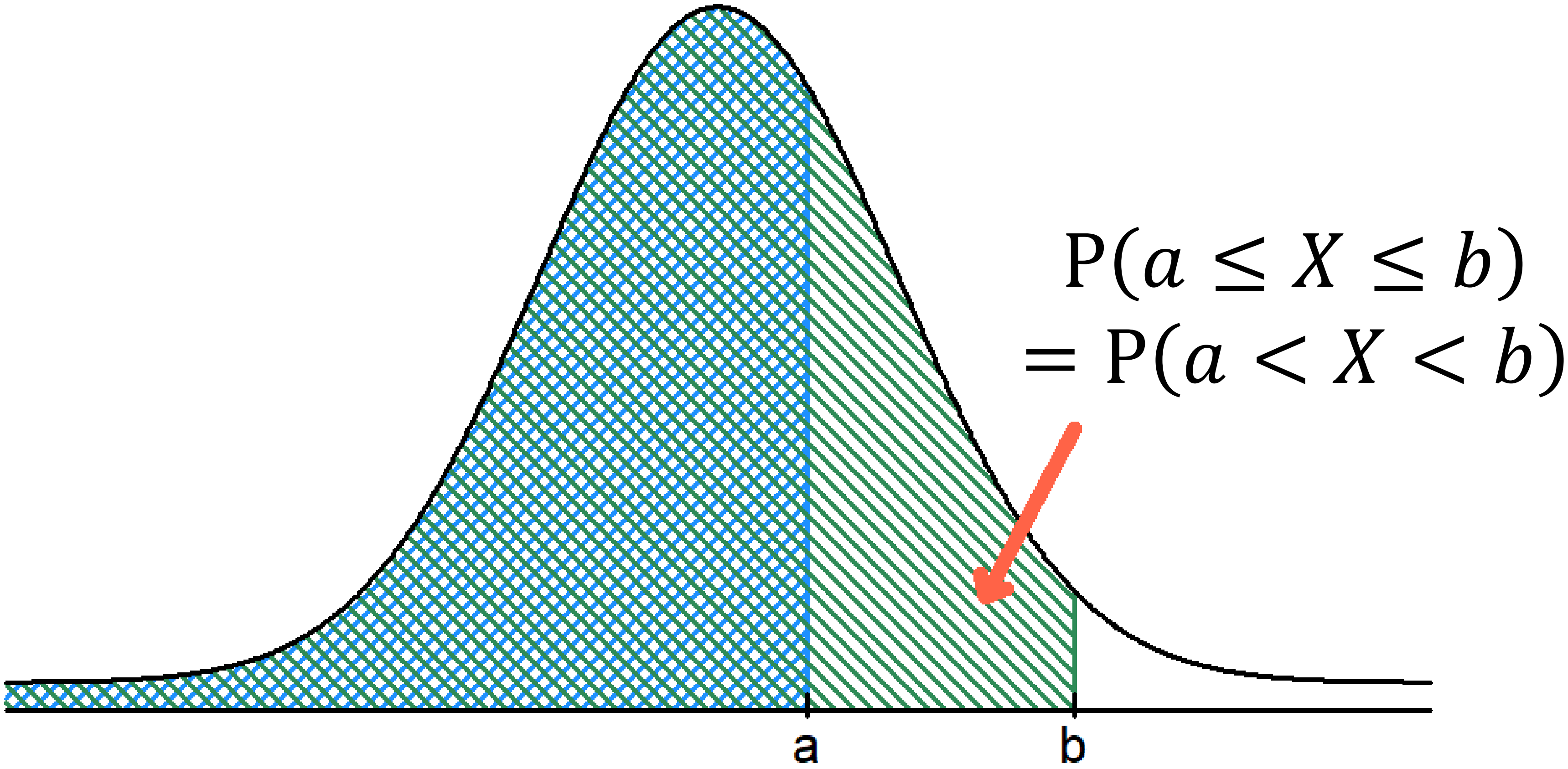
Comparing X and Z units



Note that the shape of the distribution is the same, only the scale has changed. We can express the problem in the original units of X or in standardised units (Z).

Calculate normal probabilities

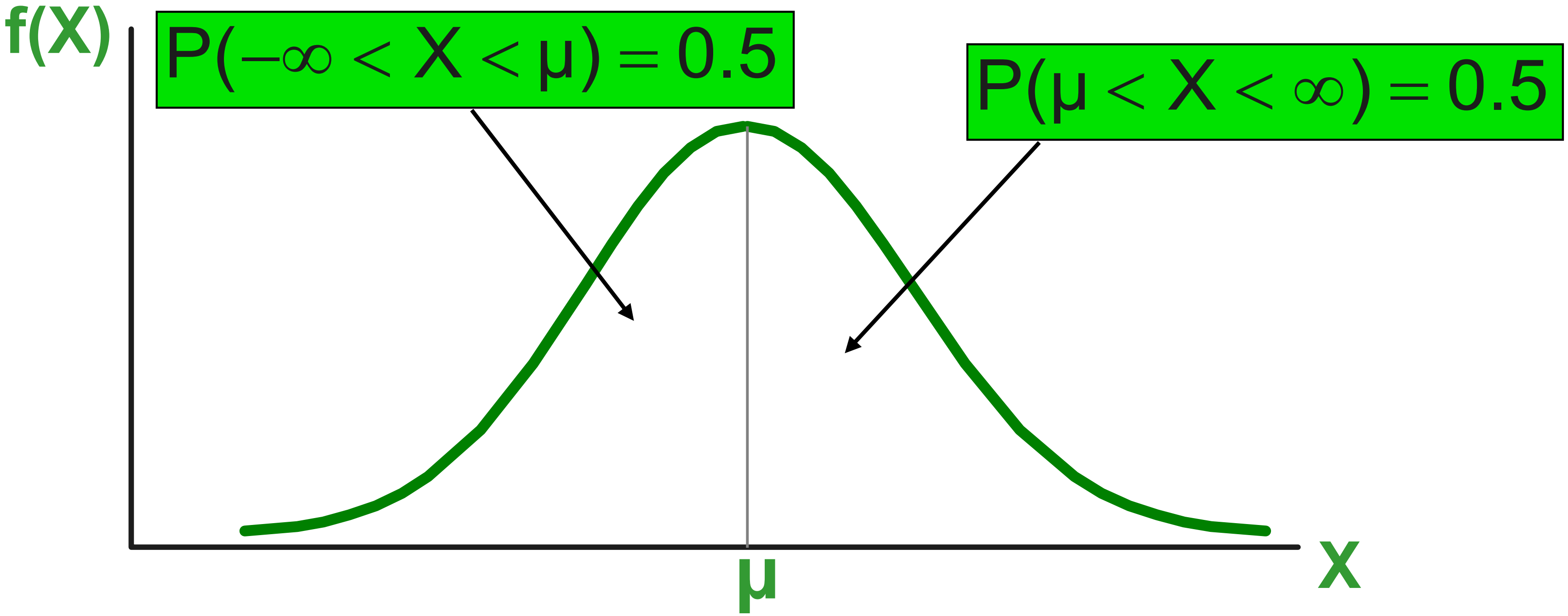
Probability is measured by the *area* under the normal curve.



Note: the probability of any individual value is zero. That is, $\mathrm{P}(X = a) = \mathrm{P}(X = b) = 0$.

Probability as area under the curve

The total area under the curve is 1, and the curve is symmetric, so half is above the mean, half is below.



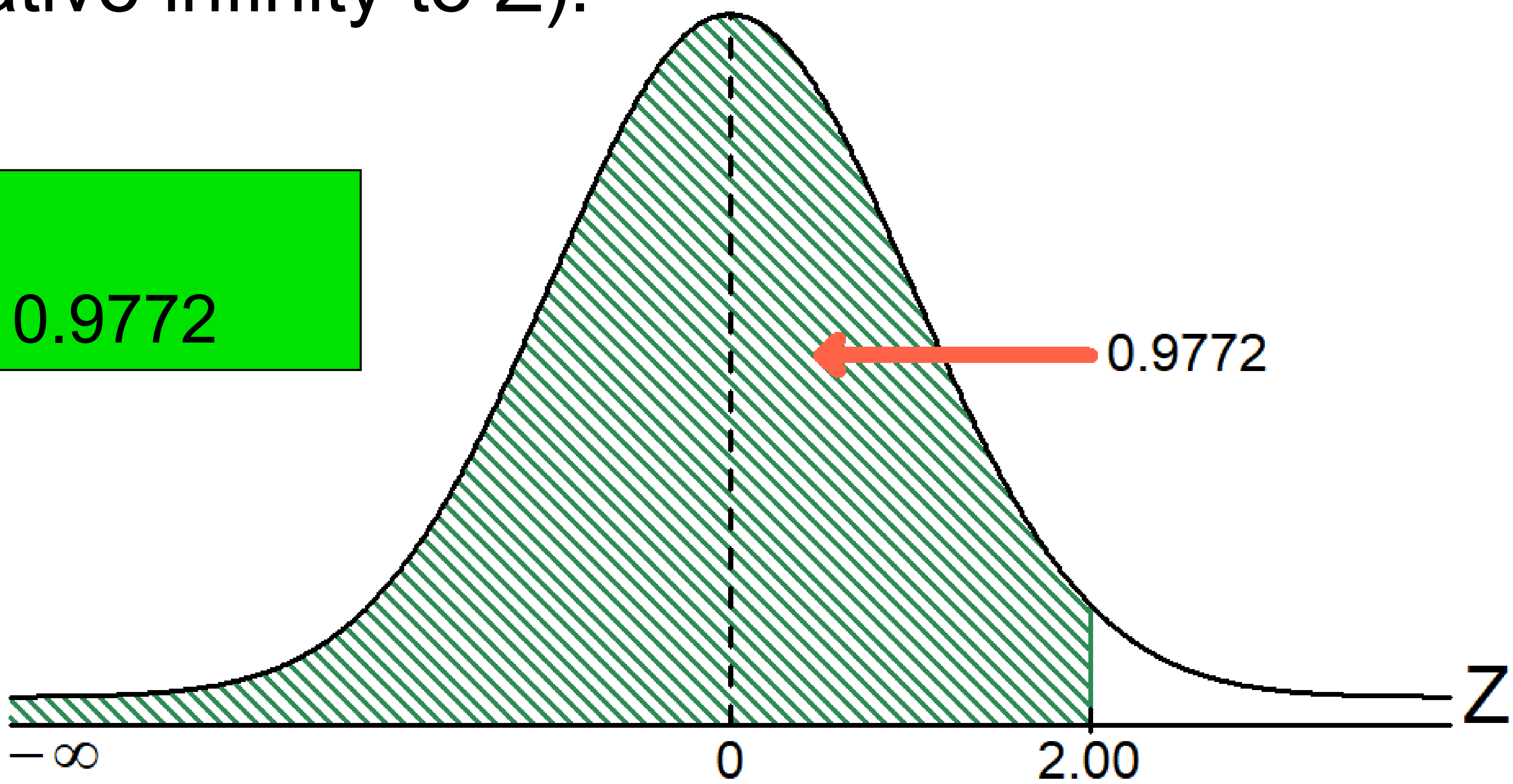
$$P(-\infty < X < \infty) = 1.0$$

The cumulative standardised normal table

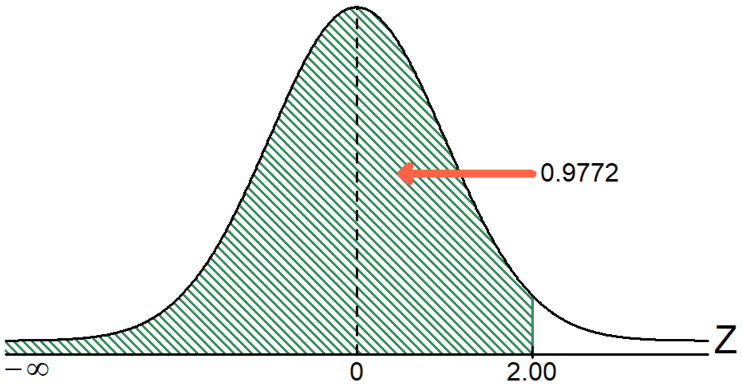
The cumulative standardised normal table gives the probability less than a desired value of Z (i.e., from negative infinity to Z).

Example:

P(Z < 2) = 0.9772



The cumulative standardised normal table (continue)



The column gives the value of Z to the second decimal point.

Z	0.00	0.01	0.02 …
0.0			
0.1			
.			
.			
.			
2.0	.9772		

The row shows the value of Z to the first decimal point.

The value within the table gives the probability from Z = – ∞ up to the desired Z value.

P(Z < 2) = 0.9772

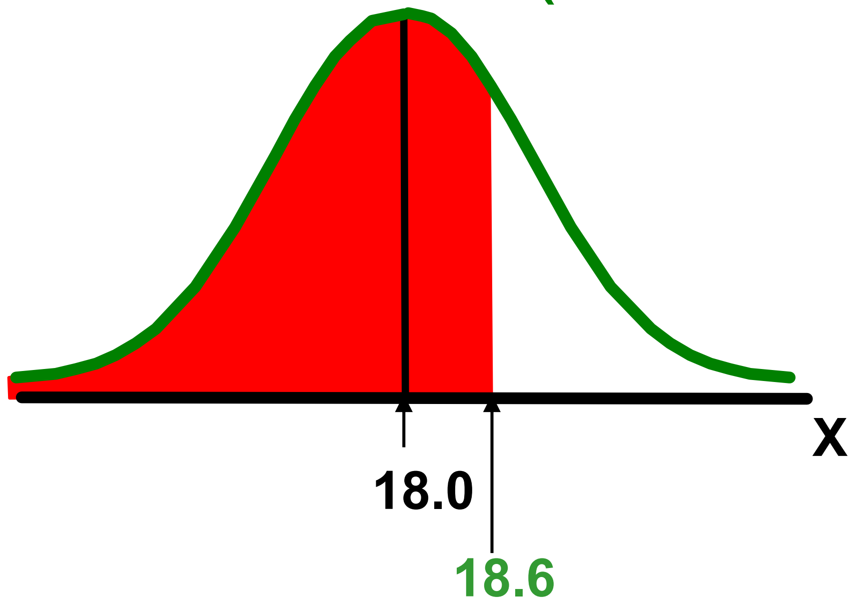
General procedure for finding normal probabilities

To calculate $P(a < X < b)$ when X is distributed normally:

- Draw the normal curve for the problem in terms of X.
- Transform X-values to Z-values.
- Use the cumulative standardised normal table.

Calculating normal probabilities

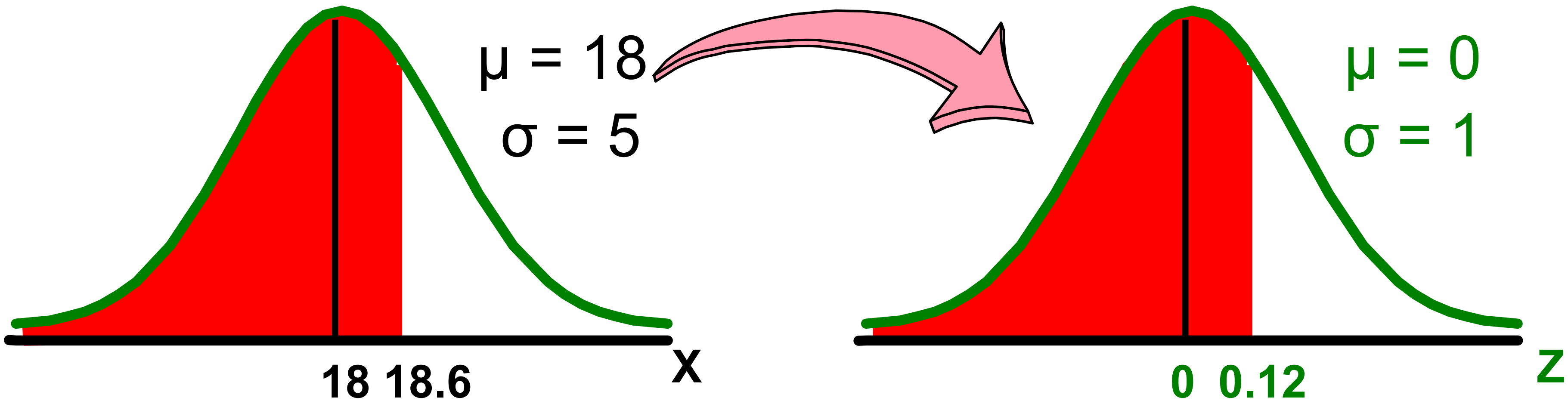
- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normally distributed with a mean of 18 seconds and a standard deviation of 5 seconds. Calculate $P(X < 18.6)$.



Calculate normal probabilities

- Let X represent the time it takes, in seconds to download an image file from the internet.
- Suppose X is normally distributed with a mean of 18 seconds and a standard deviation of 5 seconds. Calculate P(X < 18.6):

$$z = \frac{x - \mu}{\sigma} = \frac{18.6 - 18.0}{5.0} = 0.12$$



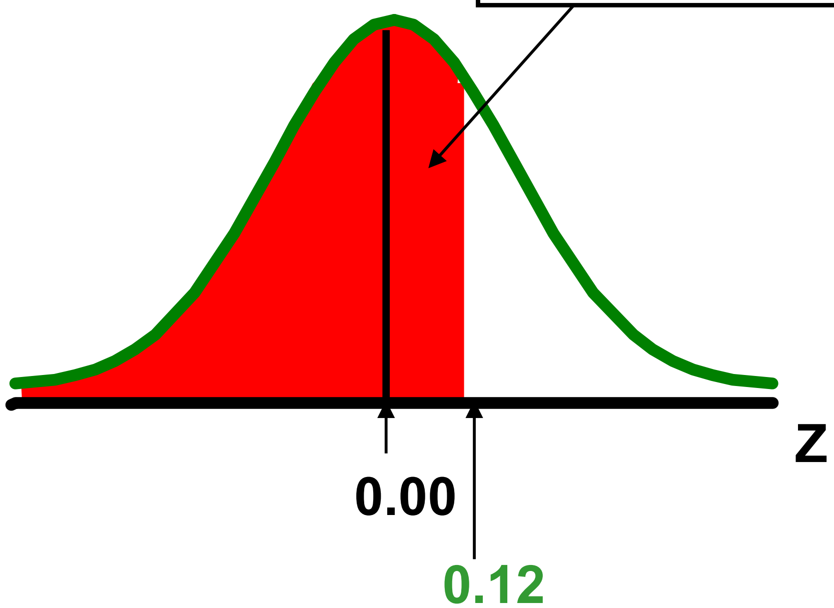
P(X < 18.6) P(Z < 0.12)

Solution: Calculate P(Z < 0.12)

Standardised normal probability table (partial)

Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

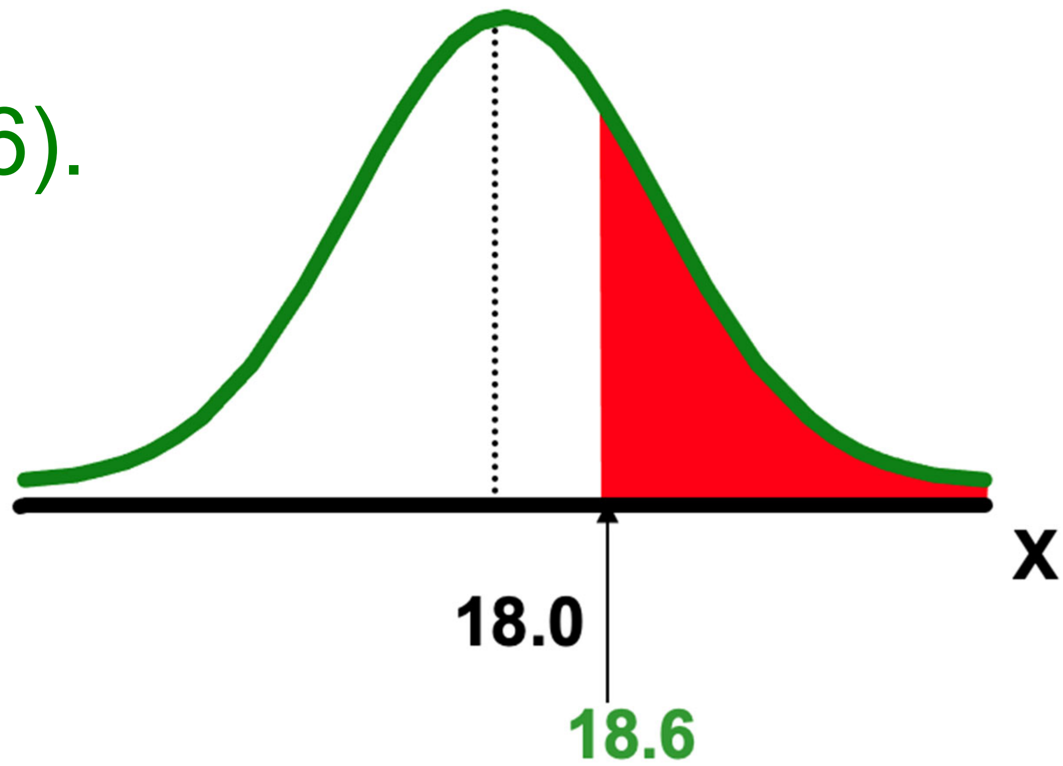
$P(X < 18.6)$
$= P(Z < 0.12)$
0.5478



Calculating normal upper tail probabilities

- Suppose X is normally distributed with mean 18 and standard deviation 5.

- Calculate P(X > 18.6).

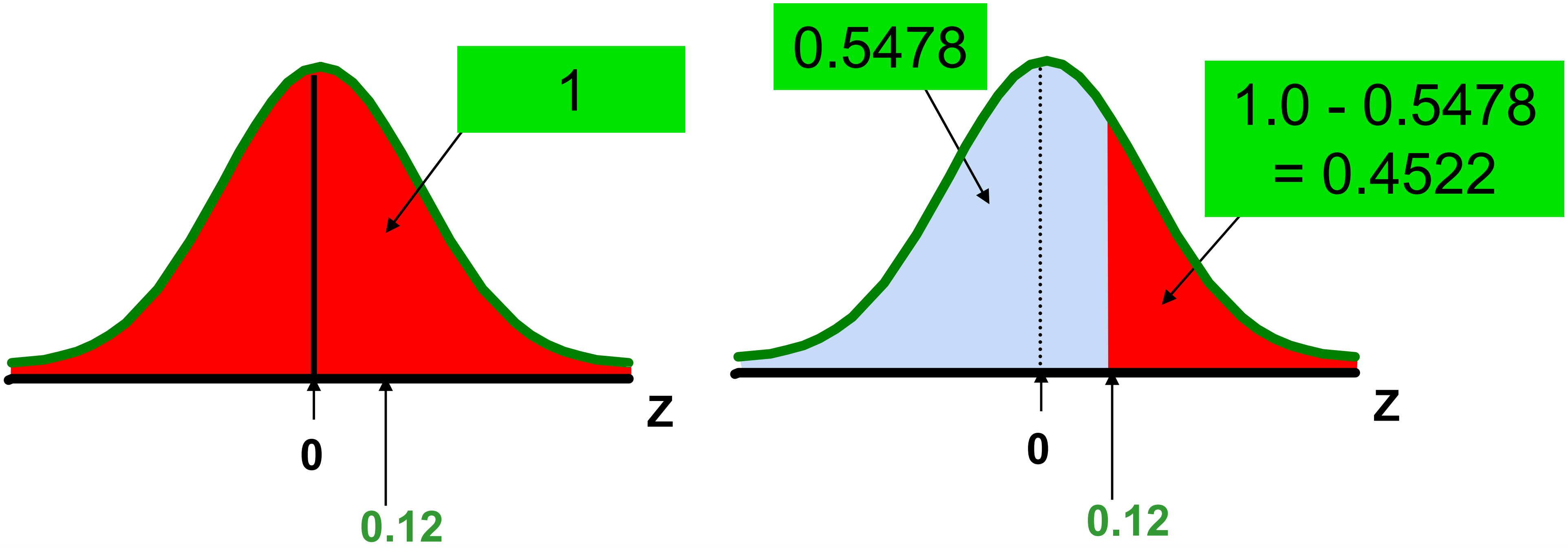


Calculate normal upper tail probabilities

- Calculate P(X > 18.6).

$$P(X > 18.6) = 1 - P(X < 18.6) = 1 - P\left(Z < \frac{18.6-18}{5}\right)$$

$$= 1 - P(Z < 0.12) = 1 - 0.5478 = 0.4522$$



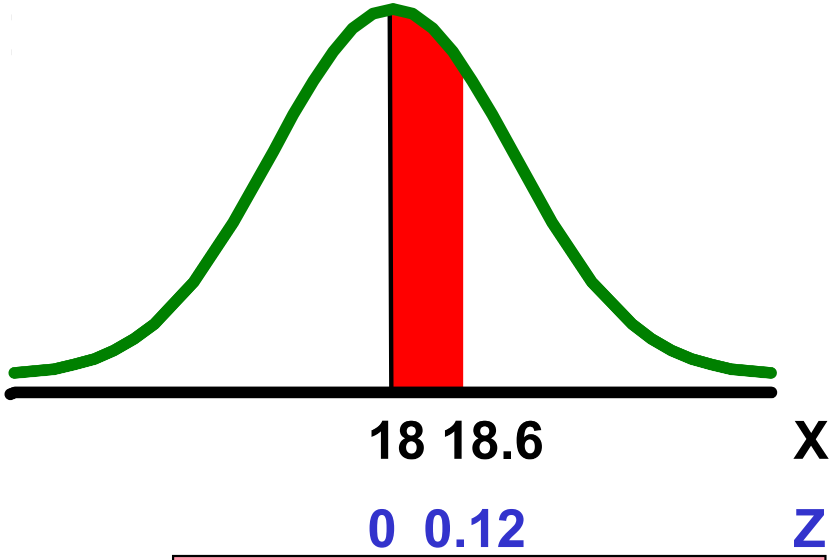
Calculating a normal probability between two X values

- Suppose X is normally distributed with mean 18 and standard deviation 5. Calculate P(18 < X < 18.6).

Calculate Z-values:

$$z = \frac{x - \mu}{\sigma} = \frac{18 - 18}{5} = 0$$

$$z = \frac{x - \mu}{\sigma} = \frac{18.6 - 18}{5} = 0.12$$



P(18 < X < 18.6)

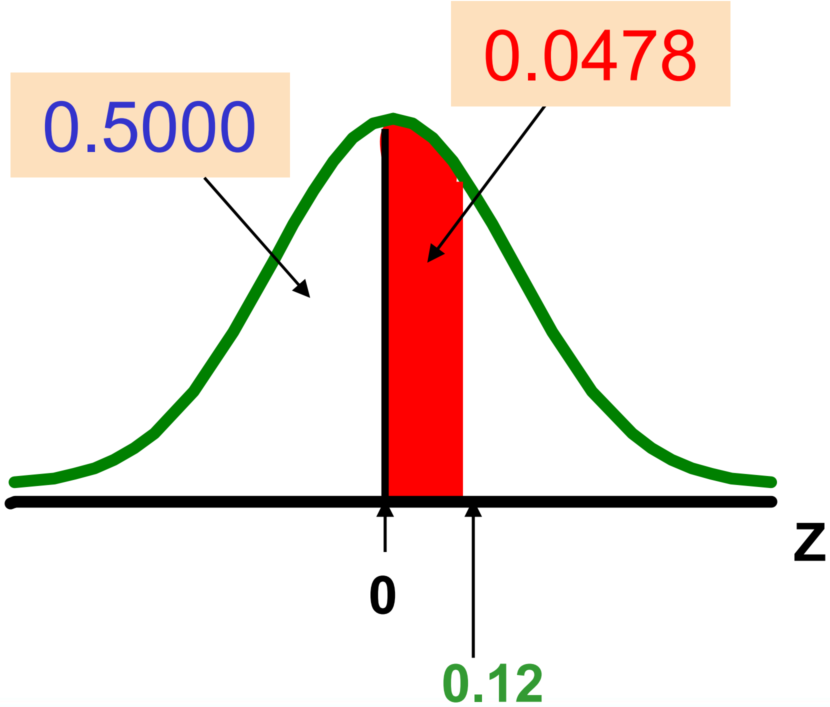
= P(0 < Z < 0.12)

Solution: Calculate P(0 < Z < 0.12)

Standardised normal probability table (partial)

Z	.00	.01	.02
0.0	.5000	.5040	.5080
0.1	.5398	.5438	.5478
0.2	.5793	.5832	.5871
0.3	.6179	.6217	.6255

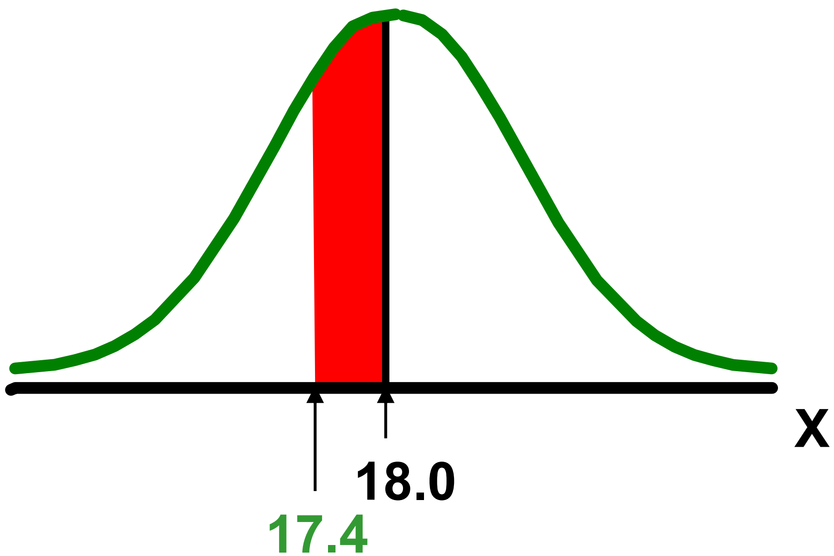
$P(18 < X < 18.6)$
$= P(0 < Z < 0.12)$
$= P(Z < 0.12) – P(Z < 0)$
$= 0.5478 - 0.5000 = 0.0478$



Probabilities in the lower tail

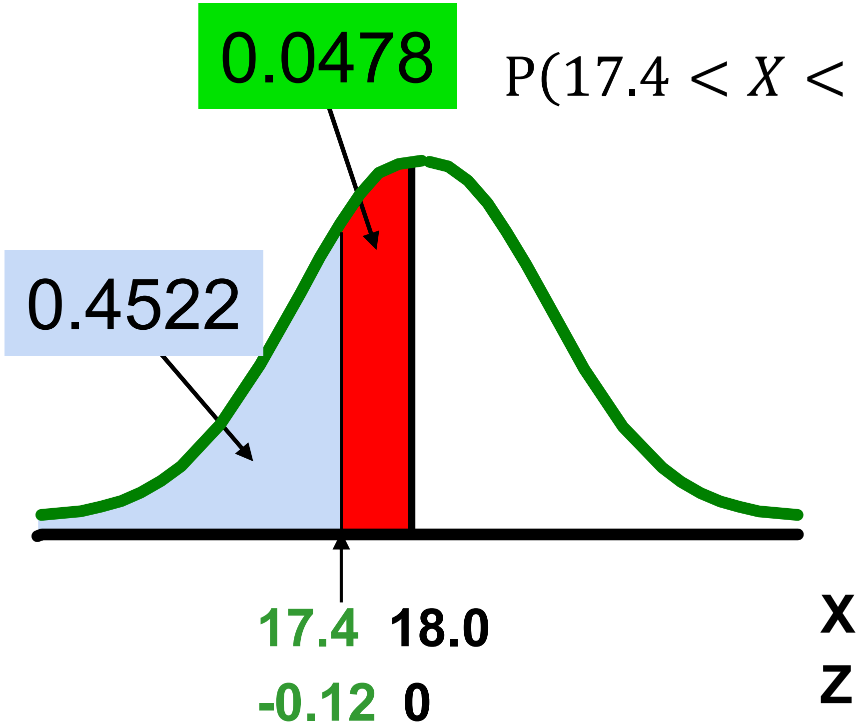
- Suppose X is normally distributed with mean 18 and standard deviation 5.

- Calculate P(17.4 < X < 18).



Probabilities in the lower tail

Calculate P(17.4 < X < 18):



$$P(17.4 < X < 18.0) = P(X < 18.0) - P(X < 17.4)$$

$$= P\left(Z < \frac{18 - 18}{5}\right) - P\left(Z < \frac{17.4 - 18.0}{5}\right)$$

$$= P(Z < 0.00) - P(Z < -0.12)$$

$$= 0.5000 - 0.4522 = 0.0478$$

The normal distribution is symmetric, so this probability is the same as P(0 < Z < 0.12).

Calculating the normal percentile: Given a normal probability, calculate the X value

- Steps to find the X value for a known probability:

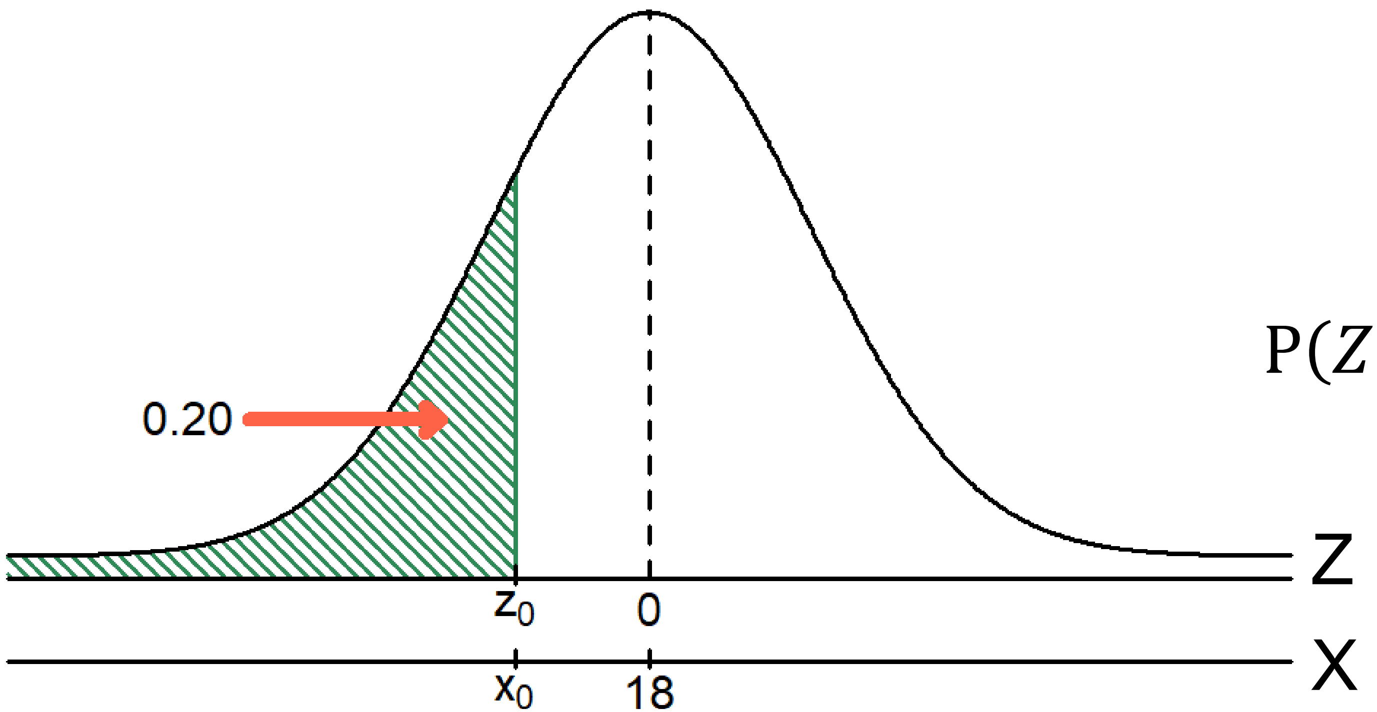
1. Find the Z value for the known probability.
2. Convert to X units using the formula:

$$\boxed{X = \mu + Z\sigma}$$

Finding the X value for a known probability

Example 1:

- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normally distributed with mean 18 and standard deviation 5.
- Find X such that 20% of download times are less than X.



$$P(Z < z_0) = 0.20$$

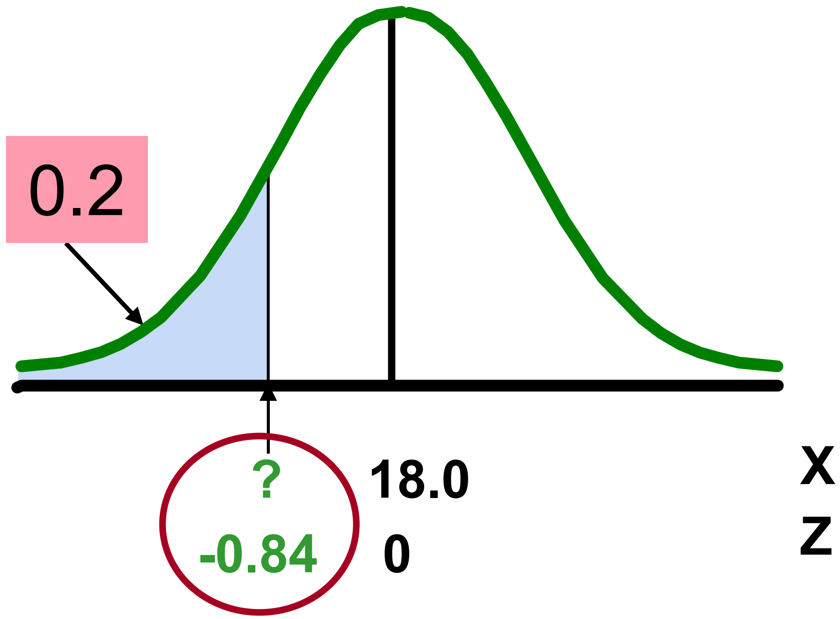
Find the Z value for 20% in the lower tail

1. Find the Z value for the known probability

Standardised normal probability table (partial)

Z	…	.03	**.04**	.05
-0.9	…	.1762	.1736	.1711
-0.8	…	.2033	.2005	.1977
-0.7	…	.2327	.2296	.2266

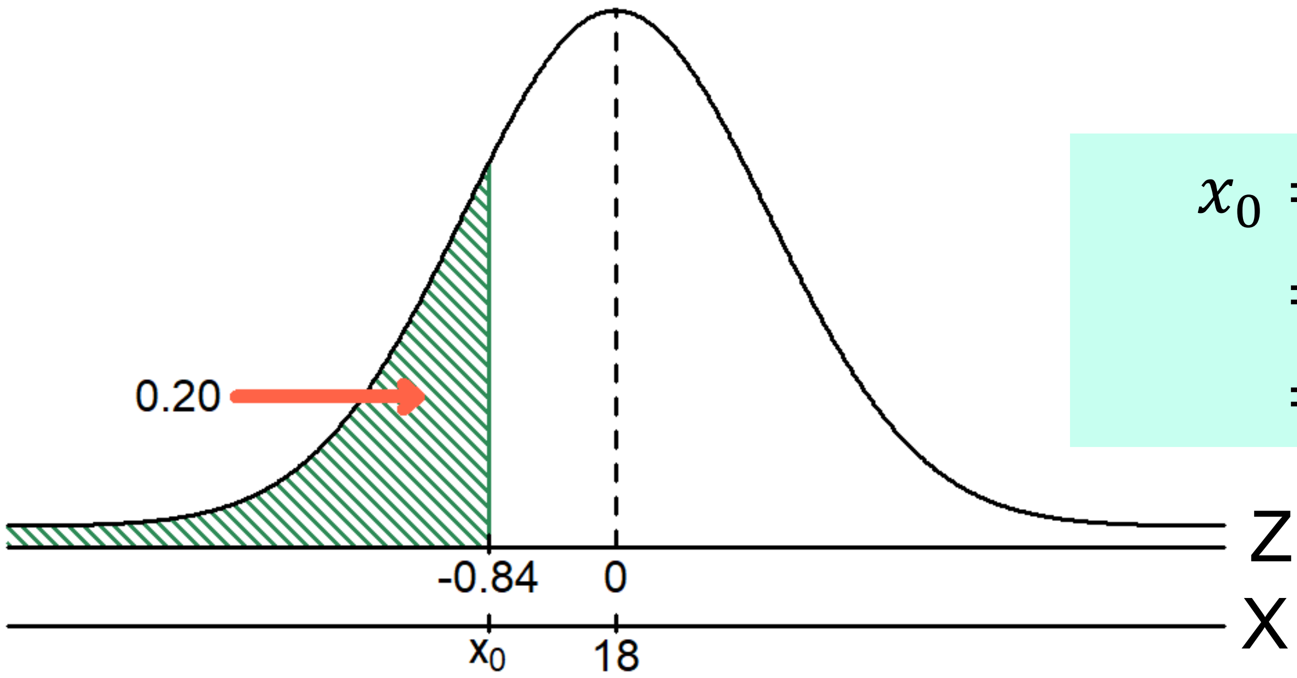
- 20% area in the lower tail is consistent with a Z value of -0.84.



Note: Use the closest probability value.

Finding the X value

2. Convert to X units using the formula:



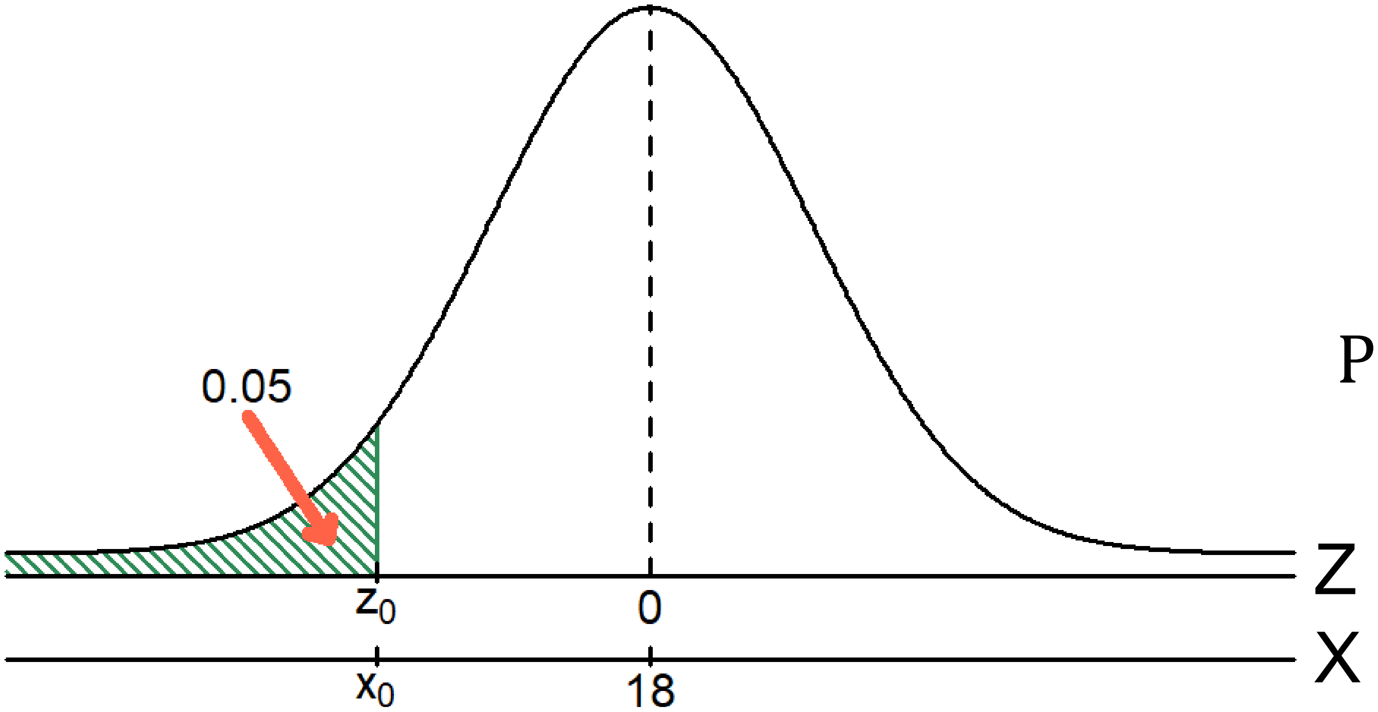
$$x_0 = \mu + z_o \times \sigma$$
$$= 18 + (-0.84) \times 5$$
$$= 13.8$$

So, 20% of the values from a distribution with mean 18 and standard deviation 5 are less than 13.80.

Finding the X value for a known probability

Example 2:

- Let X represent the time it takes (in seconds) to download an image file from the internet.
- Suppose X is normally distributed with mean 18 and standard deviation 5.
- Find X such that 5% of download times are less than X.



$$P(Z < z_0) = 0.05$$

Find the Z value for 5% in the lower tail

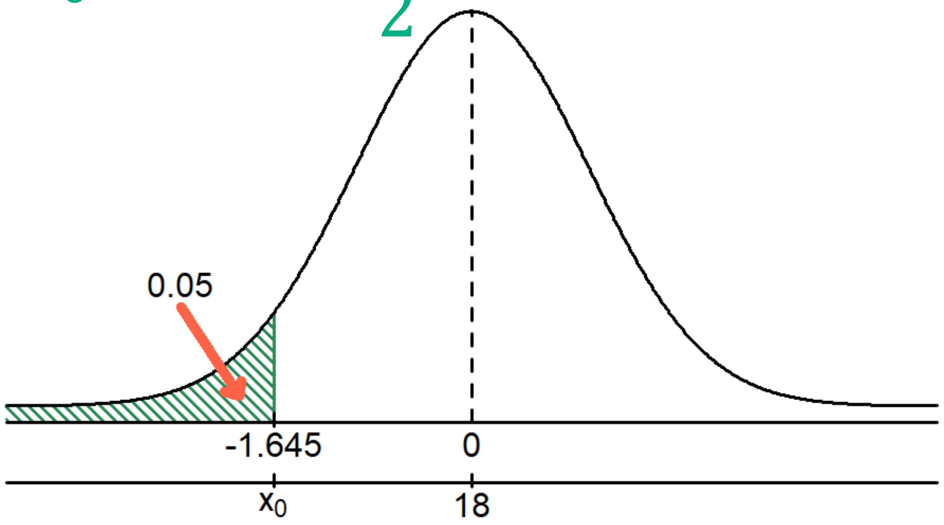
1. Find the Z value for the known probability

Standardised normal probability table (partial)

Z	…	.03	**.04**	**.05**
-1.7	…	.0418	.0409	.0401
-1.6	…	.0516	.0505	.0495
-1.5	…	.0630	.0618	.0606

- 5% area in the lower tail is consistent with a Z value:

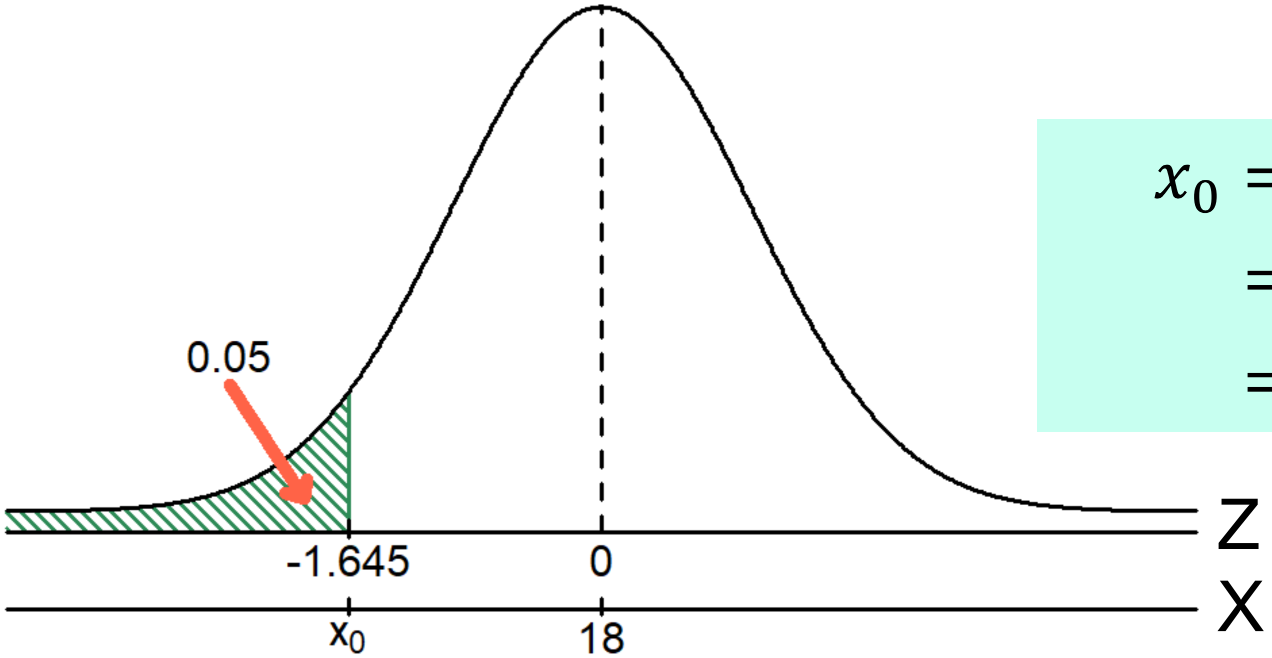
$$z_0 = \frac{-1.64 - 1.65}{2} = -1.645$$



Note: When desired probability value is in the middle, use average Z value.

Finding the X value

2. Convert to X units using the formula:



$$
\begin{aligned}
x_0 &= \mu + z_o \times \sigma \\
&= 18 + (-1.645) \times 5 \\
&= 9.7750
\end{aligned}
$$

So, 5% of the values from a distribution with mean 18 and standard deviation 5 are less than 9.78.

Excel example for computing cumulative normal probabilities

X is normally distributed with mean of 7 and standard deviation of 2:

	A	B	
1	**Normal Probabilities**		
2			
3	**Common Data**		
4	**Mean**	**7**	
5	**Standard Deviation**	**2**	
6			
7	**Probability for X <=**		
8	**X Value**	**7**	
9	**Z Value**	**0**	=STANDARDIZE(B8, B4, B5)
10	**P(X<=7)**	**0.5000**	=NORM.DIST(B8, B4, B5, TRUE)
11			
12	**Probability for X >**		
13	**X Value**	**9**	
14	**Z Value**	**1**	=STANDARDIZE(B13, B4, B5)
15	**P(X>9)**	**0.1587**	=1 - NORM.DIST(B13, B4, B5, TRUE)
16			
17	**Probability for X<7 or X >9**		
18	**P(X<7 or X >9)**	**0.6587**	=B10 + B15
19			

	A	B	
20	**Probability for a Range**		
21	From X Value	5	
22	To X Value	9	
23	Z Value for 5	-1	=STANDARDIZE(B21, B4, B5)
24	Z Value for 9	1	=STANDARDIZE(B22, B4, B5)
25	P(X<=5)	0.1587	=NORM.DIST(B21, B4, B5, TRUE)
26	P(X<=9)	0.8413	=NORM.DIST(B22, B4, B5, TRUE)
27	P(5<=X<=9)	0.6827	=ABS(B26 - B25)
28			
29	**Find X and Z Given a Cum. Pctage.**		
30	Cumulative Percentage	10.00%	
31	Z Value	-1.28	=NORM.S.INV(B30)
32	X Value	4.44	=NORM.INV(B30, B4, B5)
33			
34	**Find X Values Given a Percentage**		
35	Percentage	95.00%	
36	Z Value	-1.96	=NORM.S.INV((1 - B35)/2)
37	Lower X Value	3.08	=B4 + (B36 * B5)
38	Upper X Value	10.92	=B4 - (B36 * B5)

Evaluating data for normality

- Not all continuous variables are normal.

- It is important to evaluate how well the data set is approximated by a normal distribution.

- Normally distributed data should approximate the theoretical normal distribution:
  - The normal distribution is bell shaped (symmetrical) where the mean is equal to the median.
  - The empirical rule applies to the normal distribution.
    - Approximately 68.0% of the observations should lie within $(\mu \pm \sigma)$.
    - Approximately 95.0% of the observations should lie within $(\mu \pm 2\sigma)$.
    - Approximately 99.7% of the observations should lie within $(\mu \pm 3\sigma)$.

Evaluating data for normality

Comparing data characteristics to theoretical properties:

- Construct charts or graphs:
  - For small- or moderate-sized data sets, construct a stem-and-leaf display or a boxplot to check for symmetry.
  - For large data sets, do the histogram or polygon and assess closeness to bell-shape.
- Evaluate normal probability plot:
  - Is the normal probability plot approximately linear (that is, a straight line) with positive slope?
- Compute descriptive summary measures
  - Do the mean, median and mode have similar values?

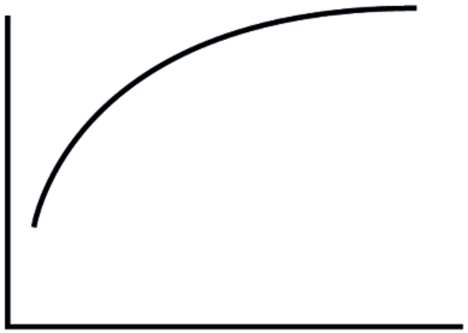
Constructing a normal probability plot

- Arrange variable data ($X = \{x_1, x_2, \dots, x_n\}$) into ordered array, where n is the number of observations in X.
- Create an index variable: $j = 1,2, \dots, n$.
- Generate probability values – $\alpha_1, \alpha_2, \dots, \alpha_n$ – such that $\alpha_j = j/(n+1)$.
- Find corresponding standardised normal quantile values (Z values).
- Plot the pairs of points with observed data values (X) on the vertical axis and the standardised normal quantile values (Z) on the horizontal axis.
- Evaluate the plot for evidence of linearity.

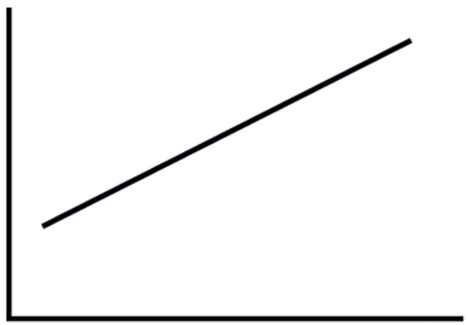
Normal probability plot interpretation

Nonlinear plots indicate a deviation from normality.

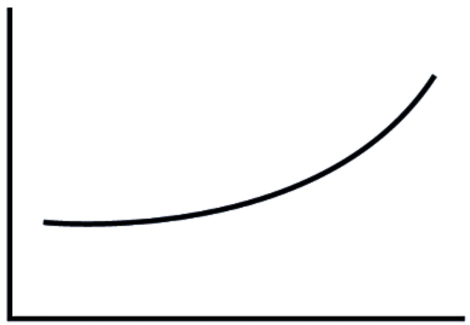
Left-skewed



Normal



Right-skewed

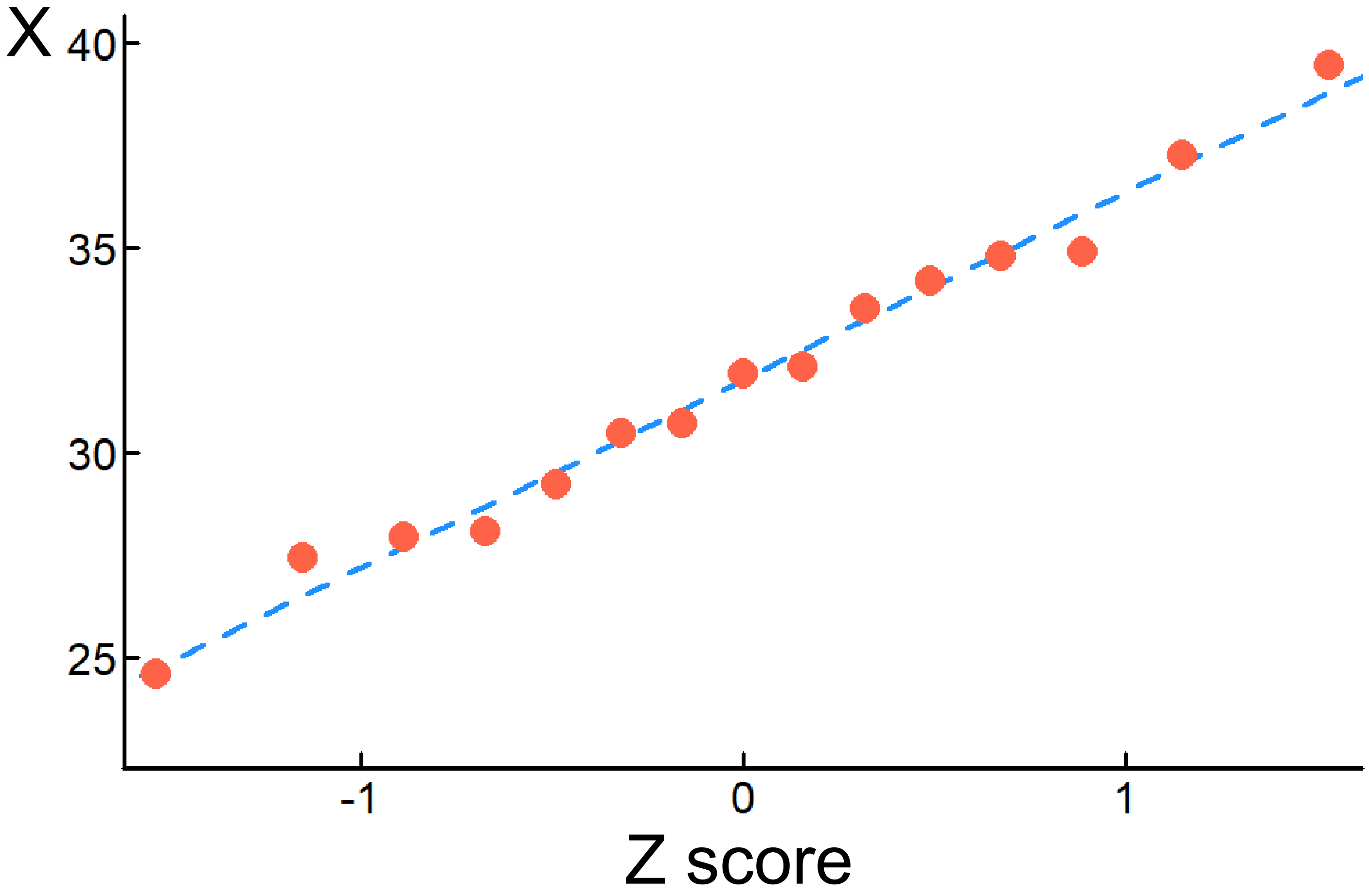


The normal probability plot interpretation

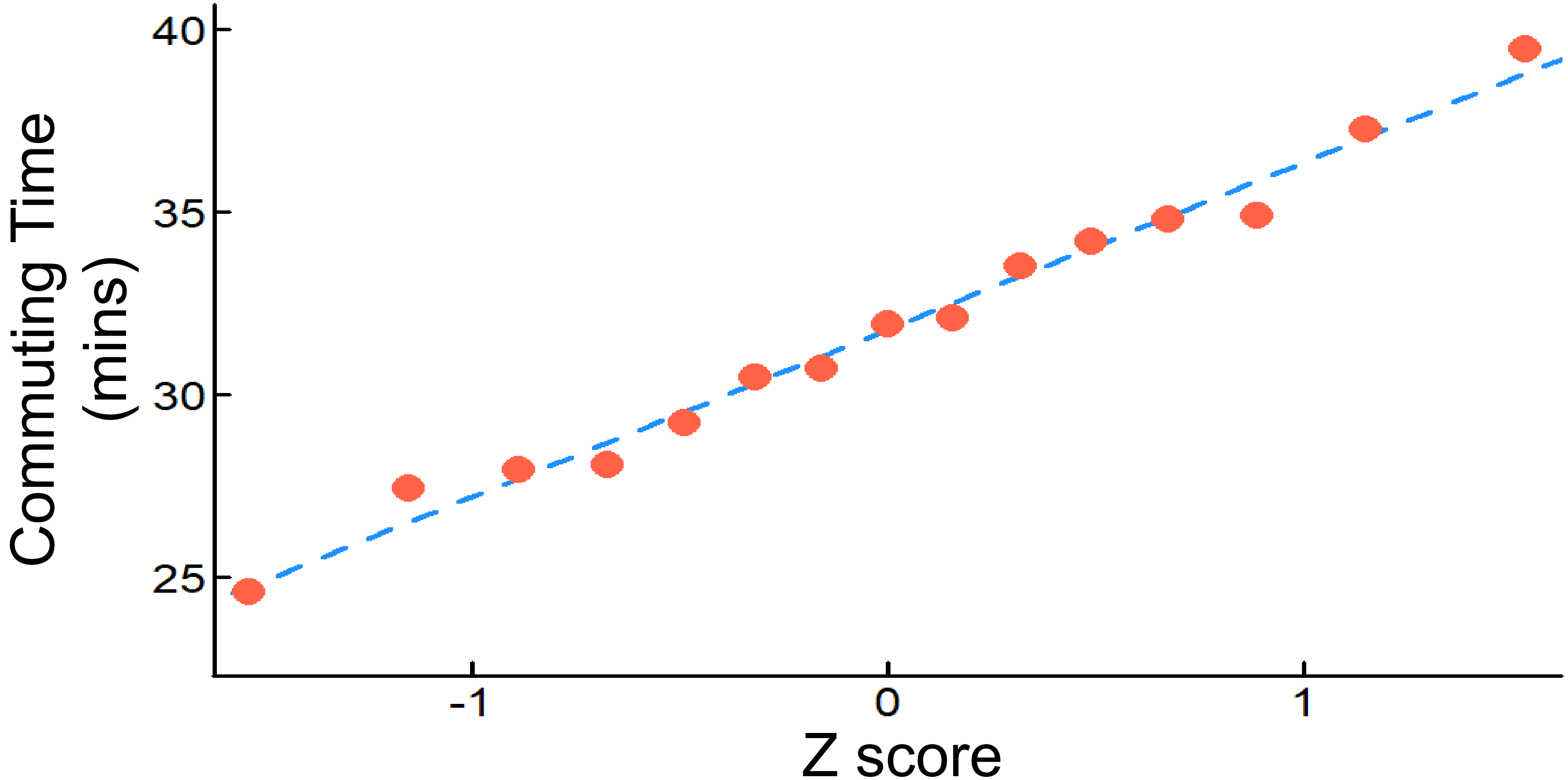
A normal probability plot for data from a normal distribution will be approximately linear.

Commuting Time (mins) Data

X	Index (j)	Probability (α)	Z Score
24.60	1	0.0625	-1.5341
27.43	2	0.1250	-1.1503
27.93	3	0.1875	-0.8871
28.08	4	0.2500	-0.6745
29.24	5	0.3125	-0.4888
30.49	6	0.3750	-0.3186
30.72	7	0.4375	-0.1573
31.93	8	0.5000	0.0000
32.11	9	0.5625	0.1573
33.51	10	0.6250	0.3186
34.19	11	0.6875	0.4888
34.79	12	0.7500	0.6745
34.91	13	0.8125	0.8871
37.26	14	0.8750	1.1503
39.47	15	0.9375	1.5341

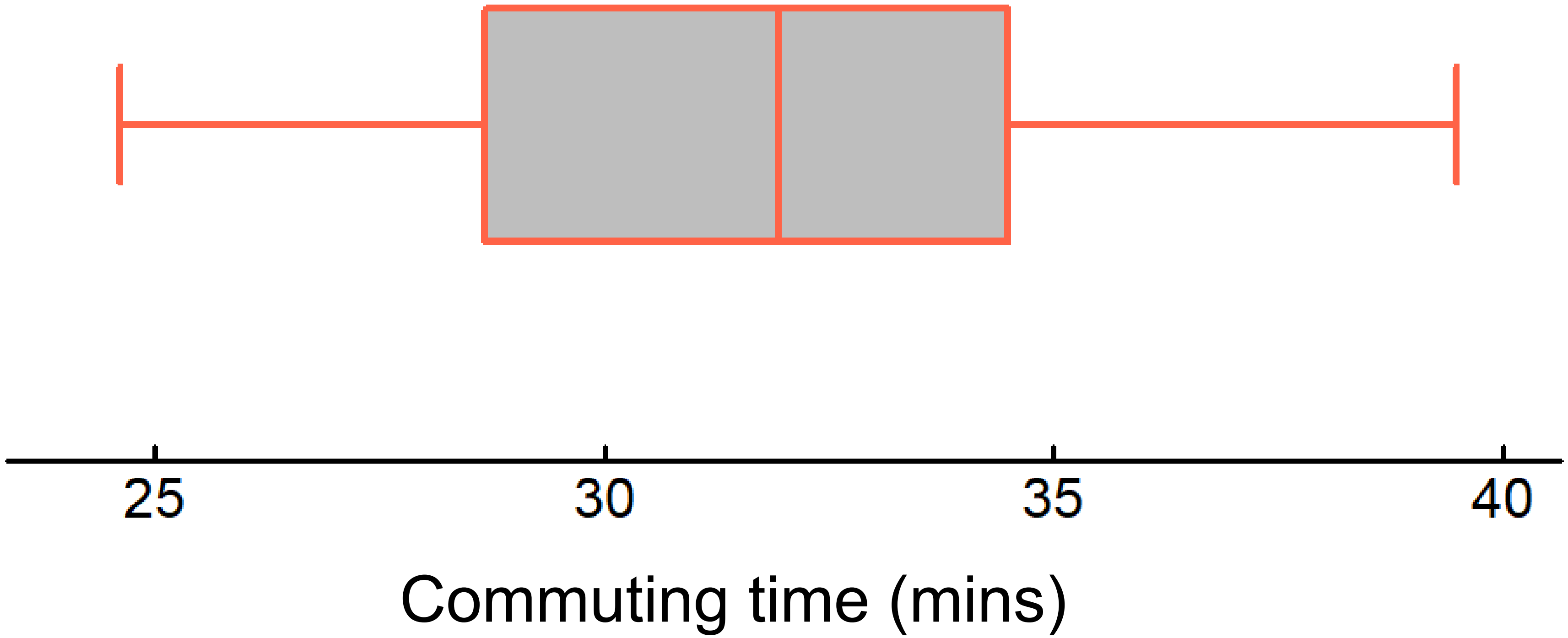


Evaluating normality: Normal probability plot



Plot is close to a straight line and shows the distribution is approximately normal. (A normal distribution appears as a straight line.)

Evaluating normality: Box plot



The box plot appears approximately symmetric.
(The normal distribution is symmetric.)

Evaluating normality:
Comparing data characteristics

Descriptive statistics for commuting time (mins) data

	A	B	C	D	E	F	G	H	I	J
1	X									
2	24.6005		Mean:		Skewness:		Q3 Position:			
3	27.4253		31.7773		0.1512		12			
4	27.9317									
5	28.0777		Median:		Kurtosis:		Q3:			
6	29.2430		31.9339		-0.3320		34.7894			
7	30.4935						Empirical Rule			
8	30.7152		Minimum:		Count:		$\mu - \sigma$	$\mu + \sigma$	Count	Proportion
9	31.9339		24.6005		15		27.7666	35.7881	11	0.7333
10	32.1101									
11	33.5144		Maximum:		Q1 position:		$\mu - 2\sigma$	$\mu + 2\sigma$		
12	34.1890		39.4738		4		23.7559	39.7988	15	1.0000
13	34.7894									
14	34.9070		Std. Dev:		Q1		$\mu - 3\sigma$	$\mu + 3\sigma$		
15	37.2554		4.0107		28.0777		19.7451	43.8095	15	1.0000
16	39.4738									

Evaluating normality:
Comparing data characteristics

- The mean time (31.78 mins) is approximately the same as the median time (31.93 mins). (In a normal distribution the mean and median are equal.)
- 73.33% of the observations are within 1 standard deviation of the mean. (In a normal distribution this percentage is $\approx 68\%$.)
- 100% of the observations are within 2 standard deviations of the mean. (In a normal distribution this percentage is $\approx 95\%$.)
- 100% of the observations are within 3 standard deviations of the mean. (In a normal distribution this percentage is $\approx 99.7\%$.)
- The skewness statistic is ≈ 0.15 and the kurtosis statistic is ≈ -0.33. These values are close to zero. (In a normal distribution, each of these statistics equals zero.)
- The difference between the median and minimum times (7.33 mins) is approximately the same as the difference between the maximum and median times (7.54 mins). (In a normal distribution these differences are equal.)
- The difference between the median and first quartile (3.85 mins) is **greater than** the difference between the third quartile and median (2.86 mins). (In a normal distribution these differences are equal.)
- The difference between the first quartile and the minimum time (3.48 mins) is **less than** the difference between the maximum time and the third quartile (4.68 mins). (In a normal distribution these differences are equal.)

Conclusion: Distribution of commuting time is approximately normal.

The uniform distribution

- The **uniform distribution** is a probability distribution that has **equal probabilities** for all possible outcomes of the random variable.

- Also called a **rectangular distribution**.

The uniform distribution

The continuous uniform distribution:

$$f(X) = \begin{cases} \dfrac{1}{b-a} & \text{if } a \le X \le b \\ 0 & \text{otherwise} \end{cases}$$

where

f(X) = value of the density function at any X value

a = minimum value of X

b = maximum value of X

Properties of the uniform distribution

- The expected value (mean) of a uniform distribution is:

$$\boxed{\mu = \frac{a+b}{2}}$$

- The standard deviation is:

$$\boxed{\sigma = \sqrt{\frac{(b-a)^2}{12}}}$$

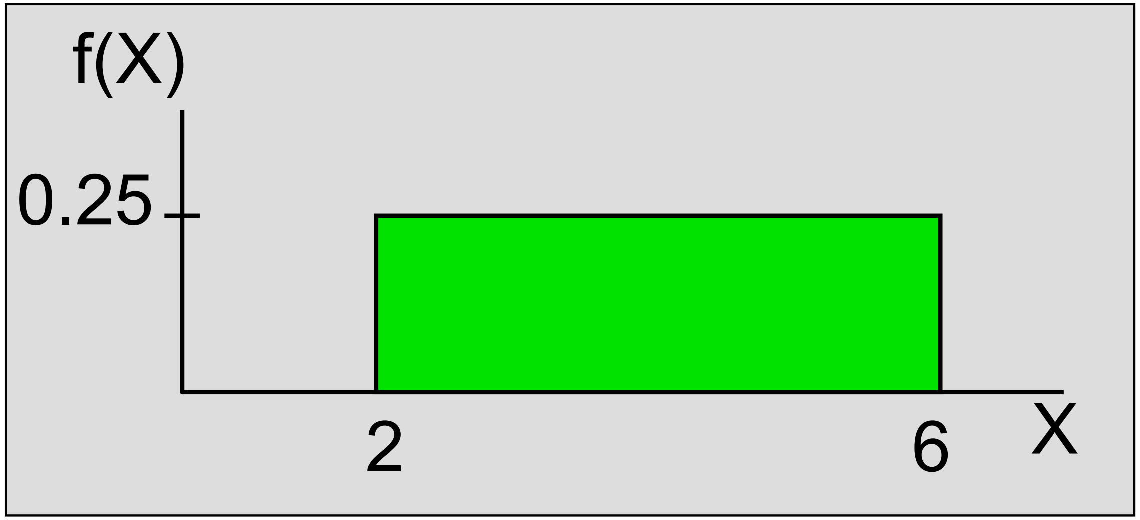
- Cumulative uniform distribution:

$$\mathrm{P}(X < t) = \frac{t-a}{b-a} \qquad \text{for } a \leq t \leq b$$

Uniform distribution example

Example: Uniform probability distribution over the range $2 \le X \le 6$:

$$f(X) = \frac{1}{6-2} = 0.25 \quad \text{for } 2 \le X \le 6$$



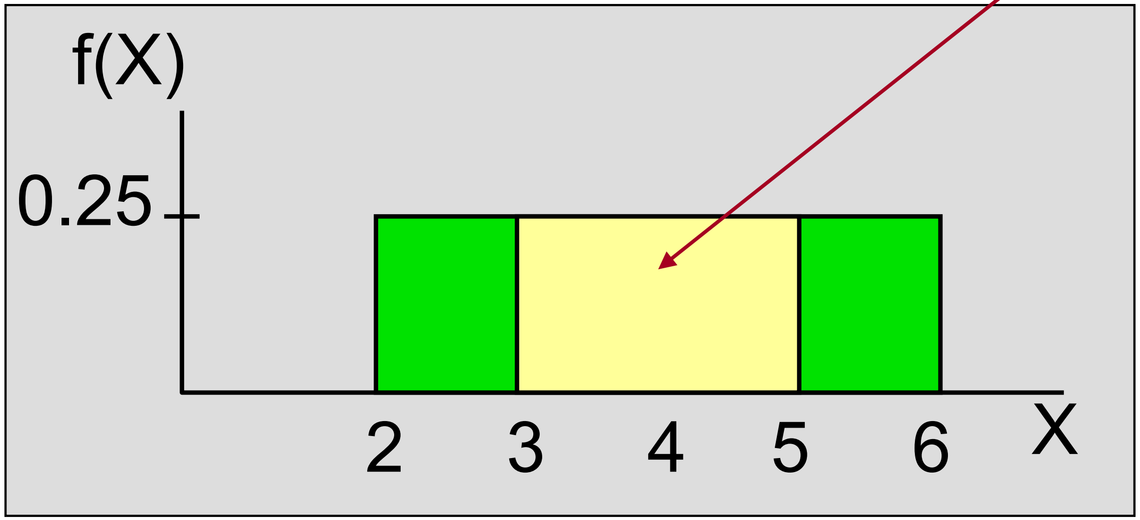
$$\mu = \frac{a+b}{2} = \frac{2+6}{2} = 4$$

$$\sigma = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(6-2)^2}{12}} = 1.1547$$

Uniform distribution example
Approach A

Example: Let X be uniformly distributed over range $2 \le X \le 6$. Calculate $P(3 \le X \le 5)$.

$$P(3 \le X \le 5) = (\text{Base})\text{x}(\text{Height}) = (2)(0.25) = 0.5$$



Uniform distribution example

Approach B

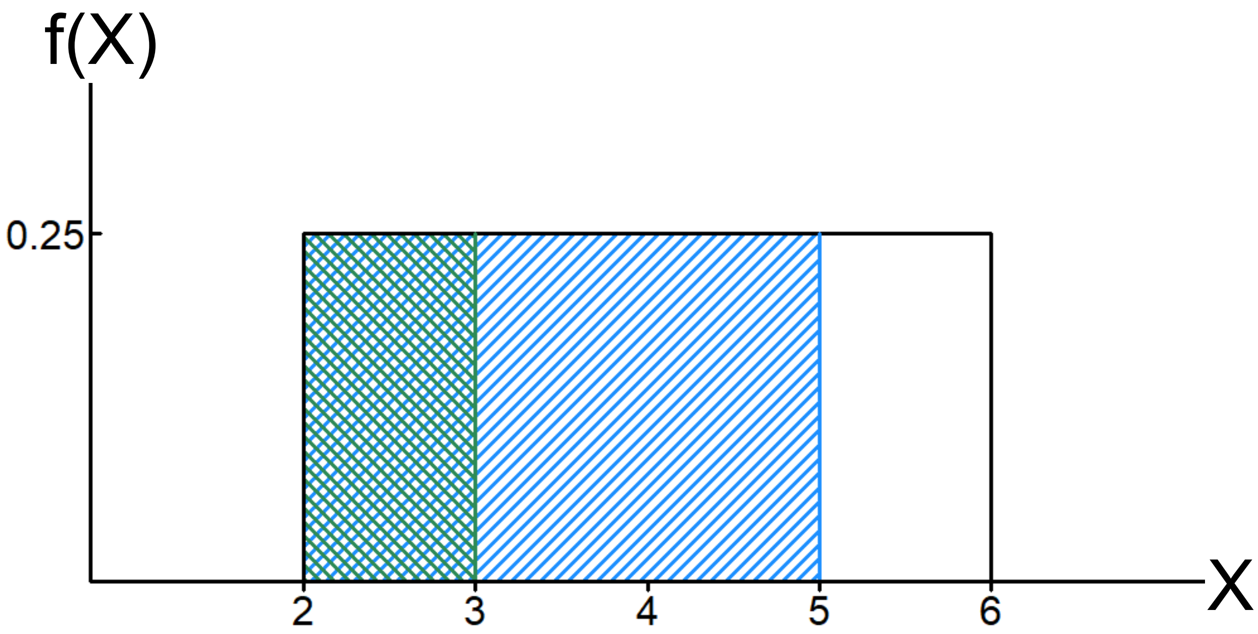
Example: Let X be uniformly distributed over range 2 ≤ X ≤ 6. Calculate P(3 ≤ X ≤ 5).

$$P(3 \le X \le 5) = P(3 < X < 5)$$

$$= P(X \le 5) - P(X \le 3)$$

$$= \frac{5-2}{6-2} - \frac{3-2}{6-2}$$

$$= \frac{3}{4} - \frac{1}{4} = 0.50$$



Normal approximation to binomial distribution

- When number of trials (n) are large, doing calculations using the **binomial** distribution may become tedious. In such case, a **normal** approximation may be used.

- If n is such that $n\pi$ and $n(1-\pi)$ are both greater than 5, then **Binomial**(n, π) may be approximated with a **Normal** $(n\pi, n\pi[1-\pi])$ distribution.

Normal approximation to binomial distribution (cont'd)

Therefore, if $X \sim \text{Binomial}(n, \pi)$ and both $n\pi$ and $n(1-\pi)$ are greater than 5, then

$$Z = \frac{X - n\pi}{\sqrt{n\pi(1-\pi)}} \sim \text{Normal}(0, 1)$$

where π is the probability of a success.

Approximate binomial estimation
Continuity correction

Binomial distribution	Normal distribution
• Only integer outcomes.	• Outcome within $(-\infty, +\infty)$.
• $\mathrm{P}(X = x) \geq 0$	• $\mathrm{P}(X = x) = 0$

- Thus, it is important to use *correction for continuity adjustment* to compute approximate binomial probabilities with normal distribution.
- Let X_a denote X approximate:
  - $X \sim Binomial(n, \pi) \quad \Longrightarrow \quad X_a \sim N(n\pi, n\pi[1-\pi])$.
  - $\mathrm{P}(X = x) \approx \mathrm{P}(x - 0.5 < X_a < x + 0.5)$.
  - $\mathrm{P}(X \leq x) \approx \mathrm{P}(X_a < x + 0.5)$.
  - $\mathrm{P}(X \geq x) \approx \mathrm{P}(X_a > x - 0.5)$.

Normal approximation to binomial distribution (Example)

A flight can accommodate 300 passengers for a single trip. However, not all bookings are usually taken up. Thus, an airline company accept 350 bookings per flight hoping that not more than 300 passengers show up for a flight. If bookings are taken up independently with a probability of 0.8, what is the probability that more than 300 passengers show up for a flight when 350 bookings have been made?

Normal approximation to binomial distribution (Example)

$$n = 350$$

$$\pi = 0.80$$

$$n\pi = 350 \times 0.80 = 280 > 5$$

$$n(1-\pi) = 350 \times (1-0.80) = 70 > 5$$

Let X represent the number of passengers that show up for a flight. Also, let X_a denote X approximate. Then,

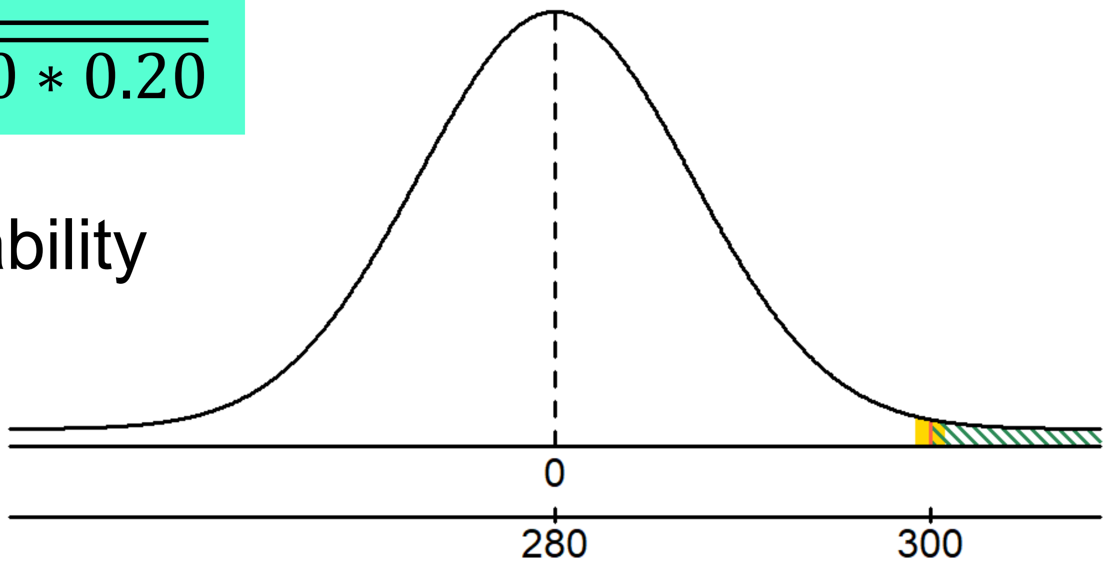
X ~Binomial$(350, 0.80)$ and $X_a \sim$ N$(280, 56)$

Approximate binomial estimation

$$z = \frac{x_a - n\pi}{\sqrt{n\pi(1-\pi)}} = \frac{x_a \; - 280}{\sqrt{350 * 0.80 * 0.20}}$$

Standardised normal probability table (partial)

Z	.02	.03	**.04**
2.6	.9956	.9957	.9959
2.7	.9967	.9968	.9969
2.8	.9976	.9977	.9977
2.9	.9982	.9983	.9984



$$\begin{aligned} P(X > 300) &= 1 - P(X \leq 300) \\ &\approx 1 - P(X_a < 300.5) \\ &= 1 - P\left(Z < \frac{300.5 - 280}{\sqrt{56}}\right) \\ &\approx 1 - P(Z < 2.74) \\ &= 1 - 0.9969 = 0.0031 \end{aligned}$$

The exponential distribution

- If events occur at random at an **average rate λ** per unit time. The **length of time** between events (X) has an exponential distribution.

- The **exponential distribution** is a continuous probability distribution that is **positively skewed** and ranges from zero to positive infinity.

The exponential distribution

The exponential density function:

$$f(X) = \begin{cases} \lambda e^{-\lambda X} & \text{if } X \geq 0, \\ 0 & \text{otherwise.} \end{cases}$$

where

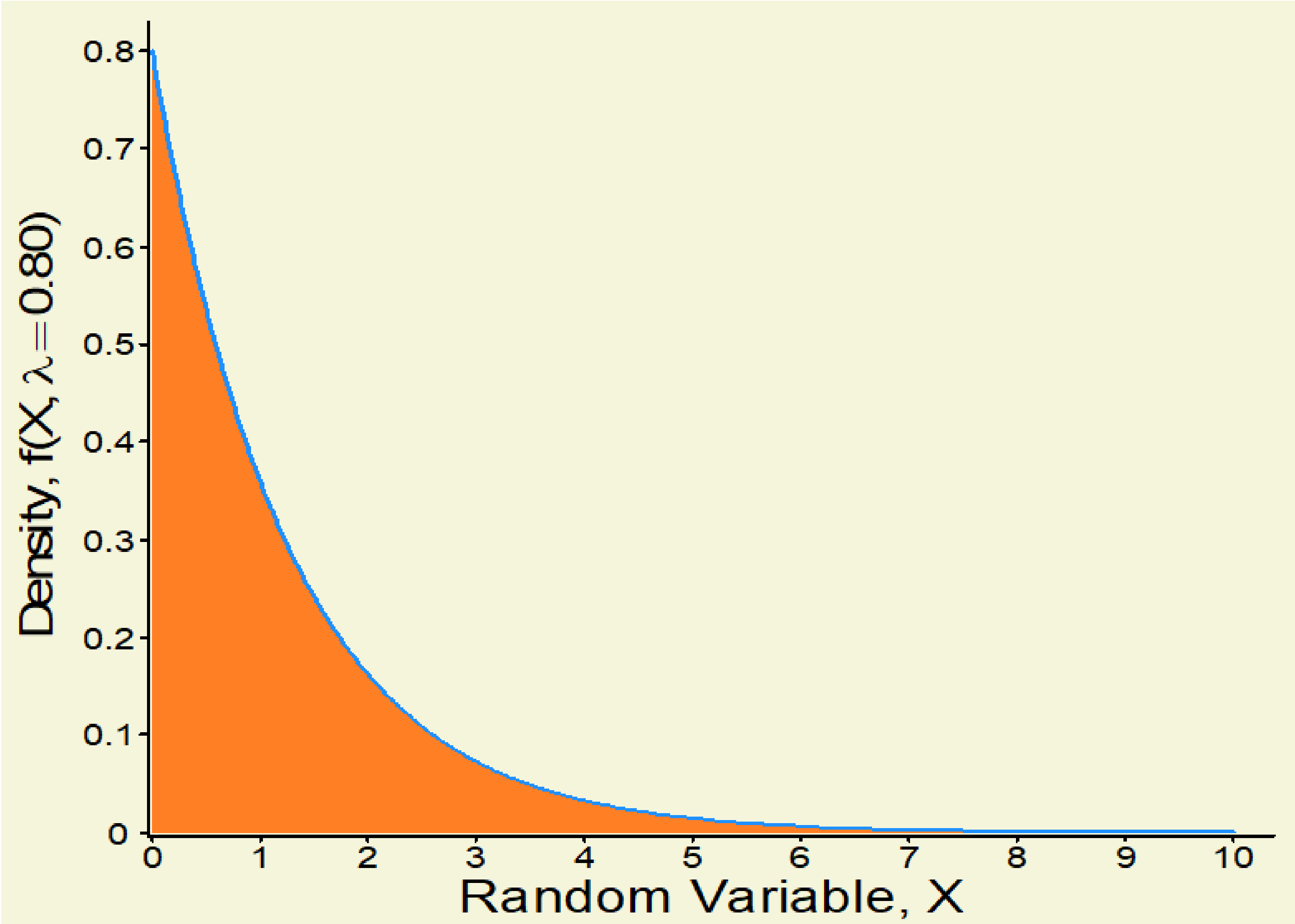
X = length of time between randomly occurring events

$f(X)$ = value of the density function at any X value

λ = average rate of event occurrence

e = Euler's Constant ≈ 2.7183

The exponential distribution



Properties of the exponential distribution

- The expected value (mean) of an exponential distribution is:

$$\mu = \frac{1}{\lambda}$$

- The variance of an exponential distribution is:

$$\sigma = \frac{1}{\lambda^2}$$

Cumulative distribution of the exponential distribution

- Probabilities may be computed for the exponential distribution with the formula:

$$P(X < t) = 1 - e^{-\lambda t} \quad \text{for} \quad 0 < t < \infty$$

where

X = length of time between randomly occurring events

λ = average rate (expected value)

e = mathematical constant approximately 2.7183

t = specific time

Exponential distribution example

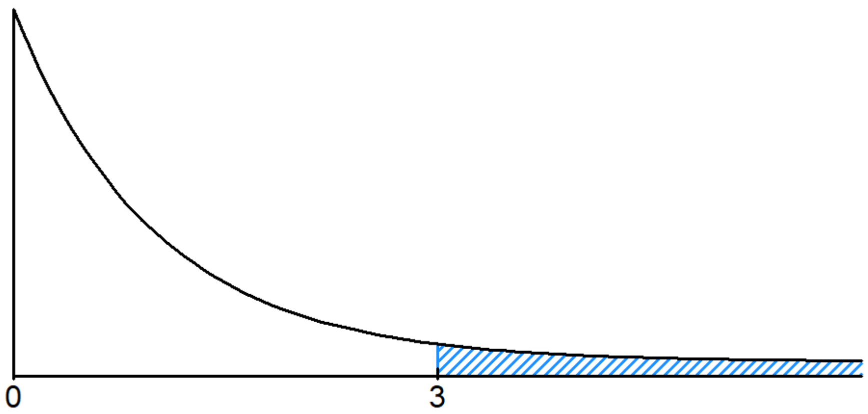
- A computer breaks down, on average, twice per week. What is the probability that the computer does not break down in a three weeks.

- Event of interest (X) is the length of time between breakdowns.

- Unit of time used to express the rate of occurrence = weekly.

- Therefore, $X \sim$ Exponential(2). Note λ =2.

Exponential distribution example



For calculation use

$$P(X < t) = 1 - e^{-\lambda t}$$

Thus, the probability that the computer will not break within **three** weeks is

$$P(X > 3) = 1 - P(X < 3) = 1 - (1 - e^{-2\times 3})$$

$$= e^{-6} \approx 0.0025$$

Chapter summary

In this chapter we discussed:

- Computing probabilities from the normal distribution.
- Using the normal distribution to solve business problems.
- Using the normal probability plot to determine whether a set of data is approximately normally distributed.
- Compute probabilities from the uniform distribution.
- Approximate binomial with normal distribution.
- Compute probabilities from the exponential distribution.